Development of the Respiratory System

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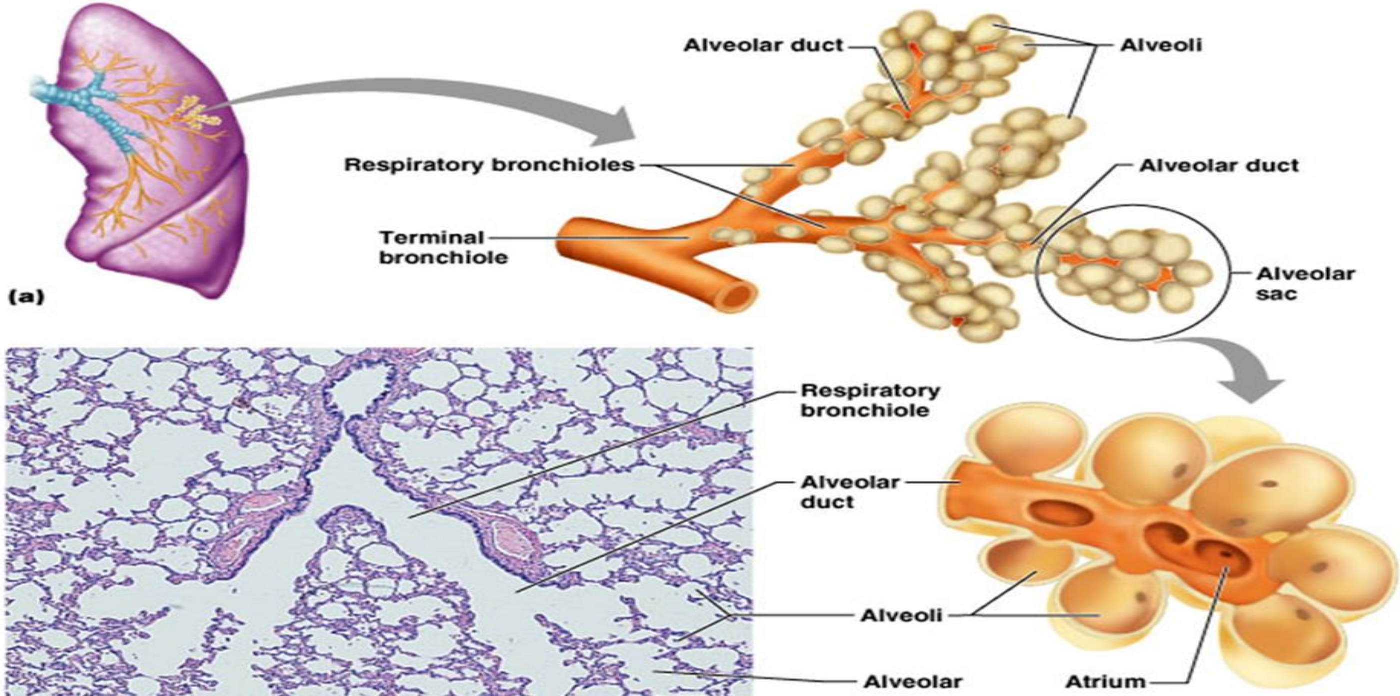
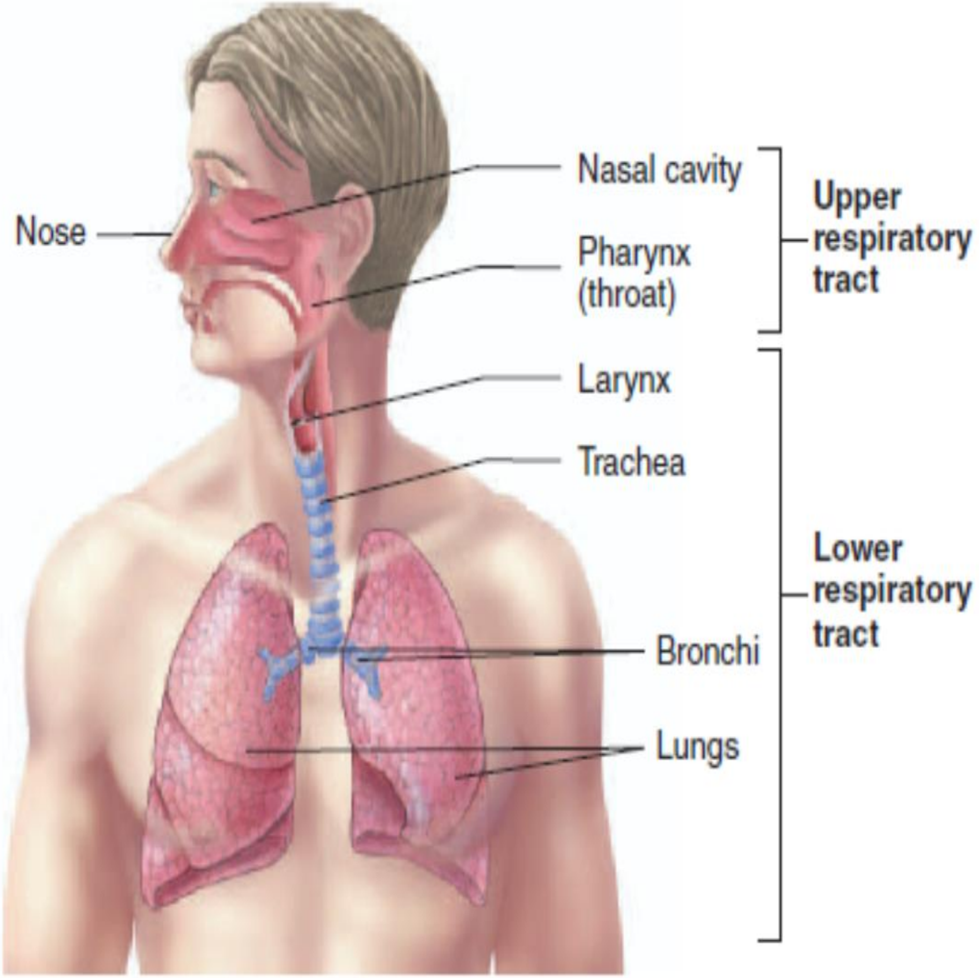
- LEARNING OBJECTIVES
- • At the end of the lecture, students should be able to:
- 1. Define the series of events that culminates in the formation of foetal respiratory organs
- 2. Define the stages of lung maturation
- 3. Understand the basis of congenital anatomic anomalies of respiratory organs
- 4. Understand the developmental basis of respiratory distress syndrome (Hyaline membrane disease)

- Respiratory system -consists of the nose, nasal cavity, pharynx , larynx , trachea, bronchi & lungs
- *Subdivisions of respiratory system*:
- 1—— *Anatomically* divided into upper & lower respiratory tracts
- ***Upper respiratory tract*** *:-*
- External nose, nasal cavities, paranasal air sinuses, pharynx
- ***Lower respiratory tract*** *:-*
- larynx, trachea, bronchi, terminal bronchioles, respiratory bronchioles & alveoli

- 2—— *Functionally* divided into conducting part & respiratory part
- ***Conducting part /zone***:-
- entire upper respiratory tract & lower respiratory part up to terminal bronchioles
- facilitates conduction, cleaning, warming & moistening of air
- ***Respiratory part/zone:-***
- respiratory bronchioles, alveolar ducts & alveoli
- concerned with exchange of gases





- ***Developmental primordia of respiratory system*:**
- *endodermal* in origin
- develops from:-

❖ a median *endodermal* diverticulum of the foregut (*respiratory diverticulum*)

❖ *splanchnopleuric intraembryonic mesoderm*

- lining epithelium of larynx, trachea, bronchi, bronchioles & alveoli are *endodermal* in origin
- cartilages, muscles & connective tissue components develop from **splanchnopleuric mesoderm** surrounding the foregut

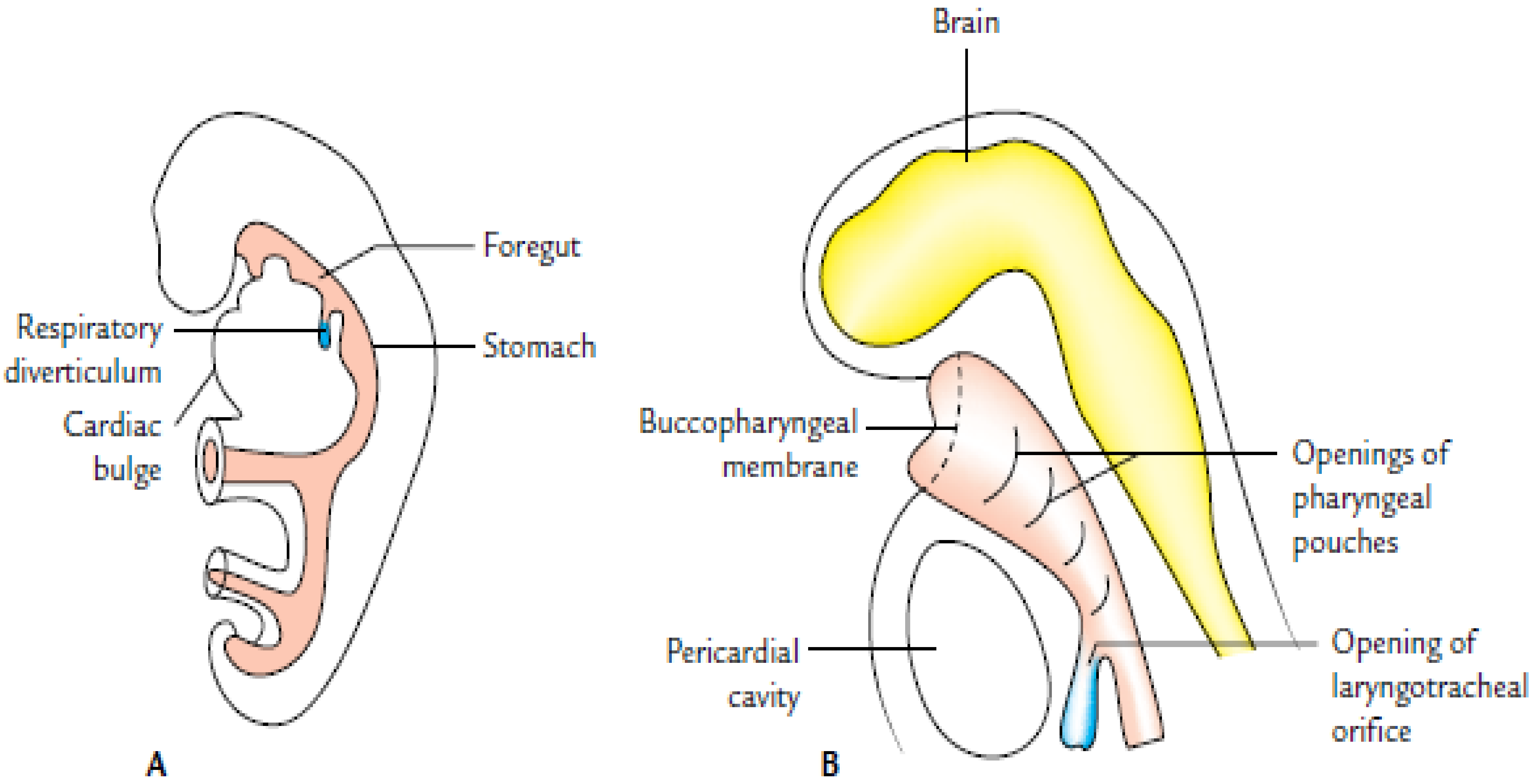


Fig. 16.1 Site of respiratory diverticulum as seen in embryo. A. A 25-day-old embryo. Note the relationship of diverticulum to stomach. B. Sagittal section of 5-week-old embryo showing openings of pharyngeal pouches and laryngotracheal orifice.

- **Development of respiratory (laryngotracheal) diverticulum:**
- appears 4th week as midline evagination known as ***respiratory diverticulum*** **or** ***laryngotracheal diverticulum*** from the ventral wall of pharyngeal part of foregut
- caudal to hypobranchial eminence (epiglottis)
- in the floor of developing pharynx
- extends in caudal direction & elongates
- Induction factors:-
- Retinoic acid, and TBX4 transcription factor
- From surrounding mesoderm

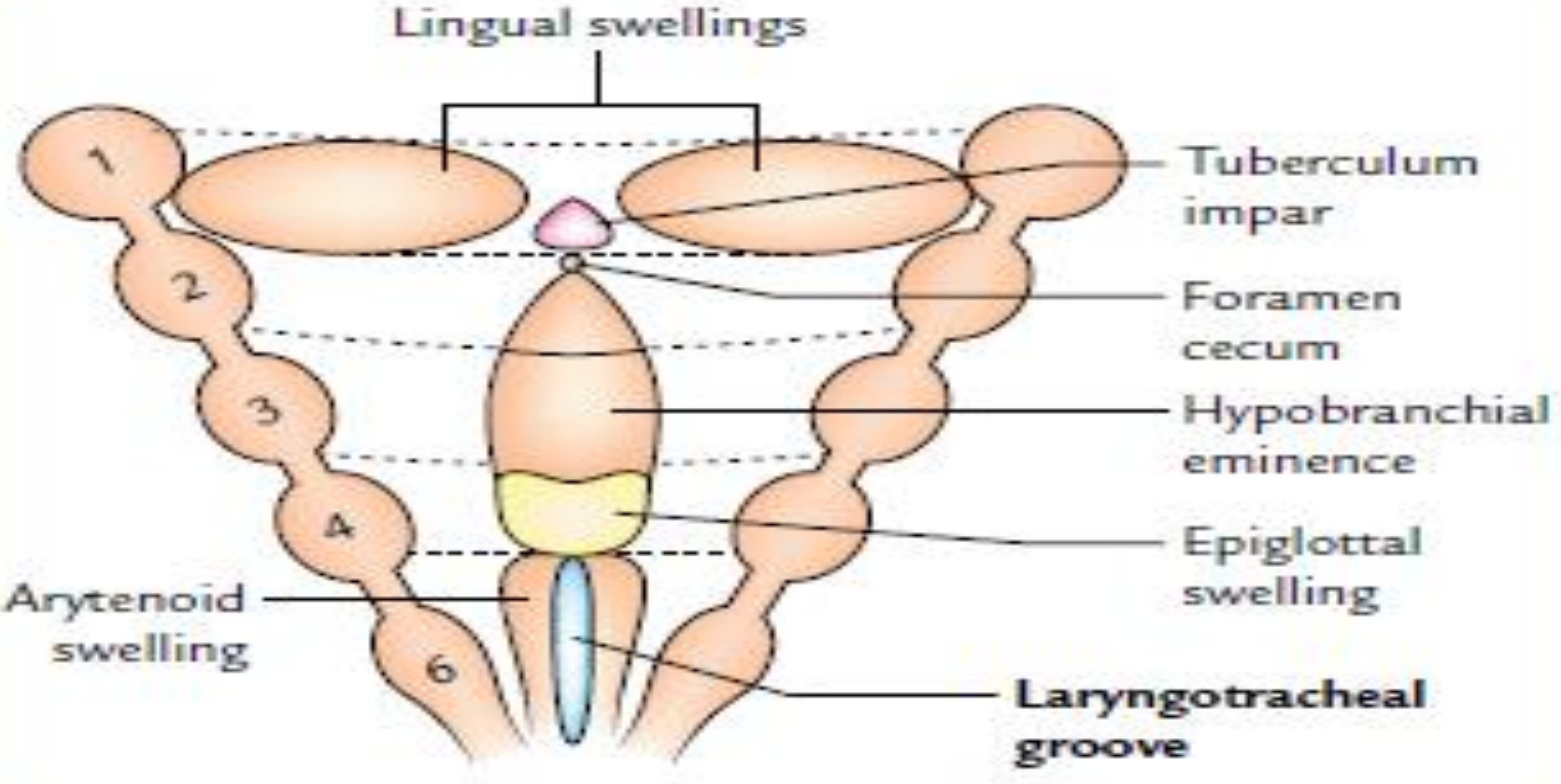
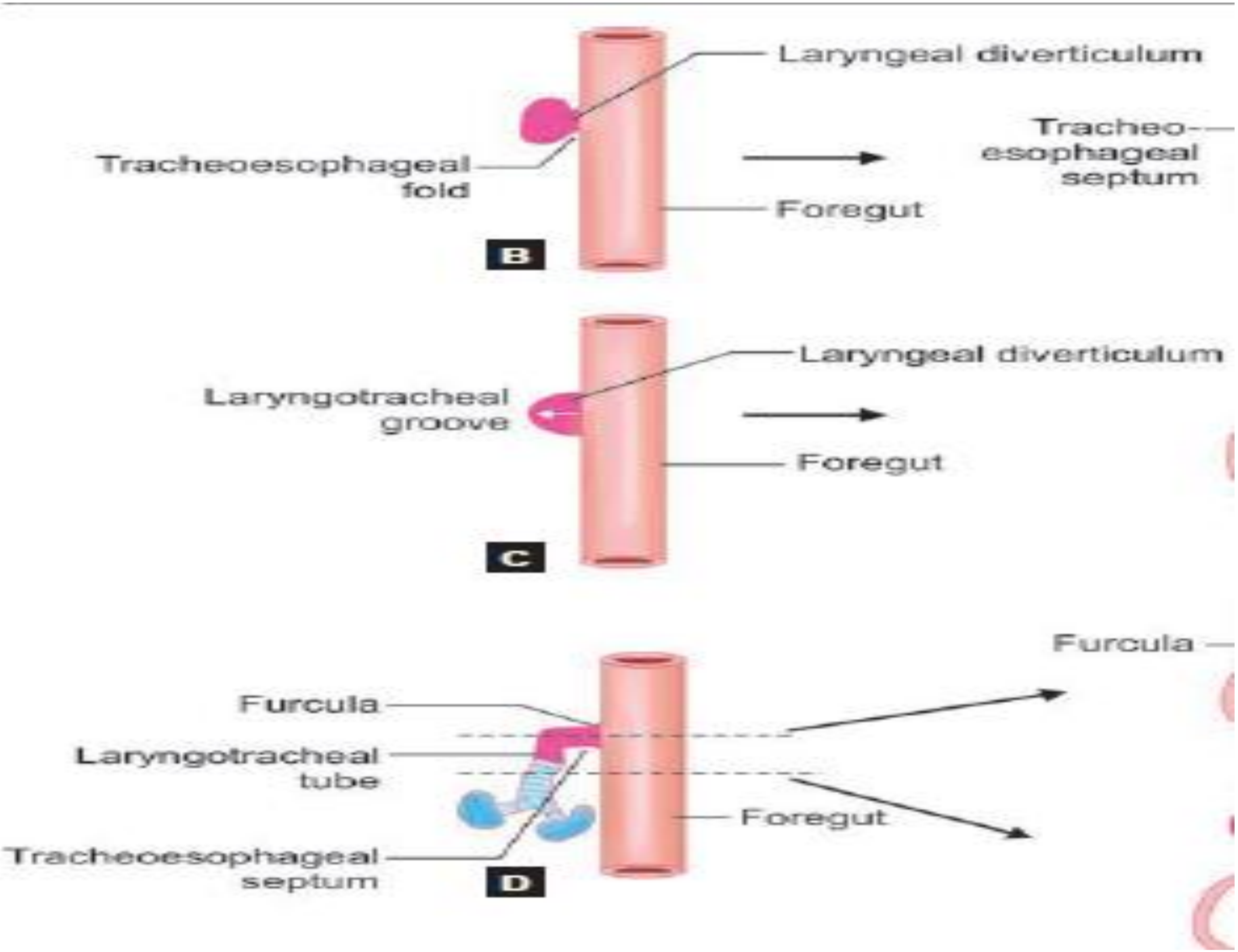
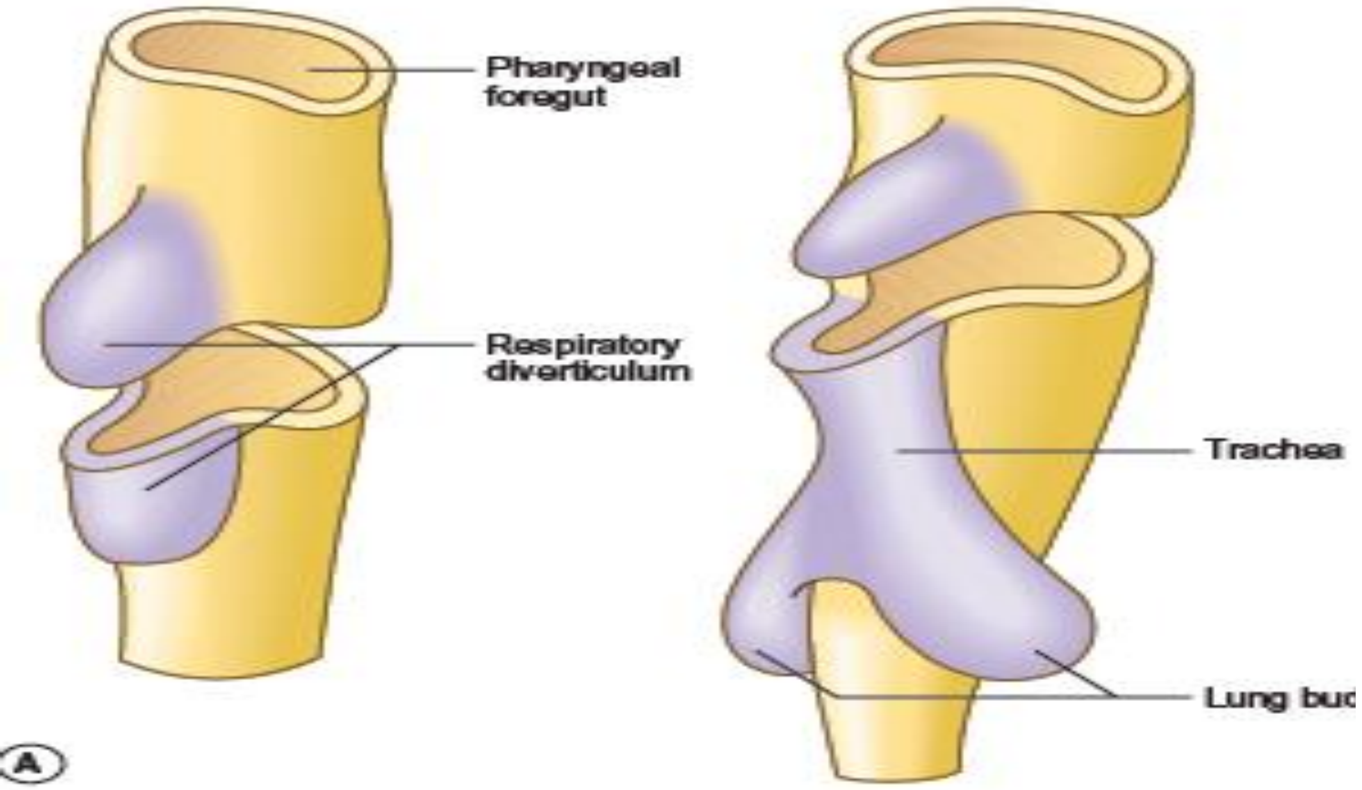


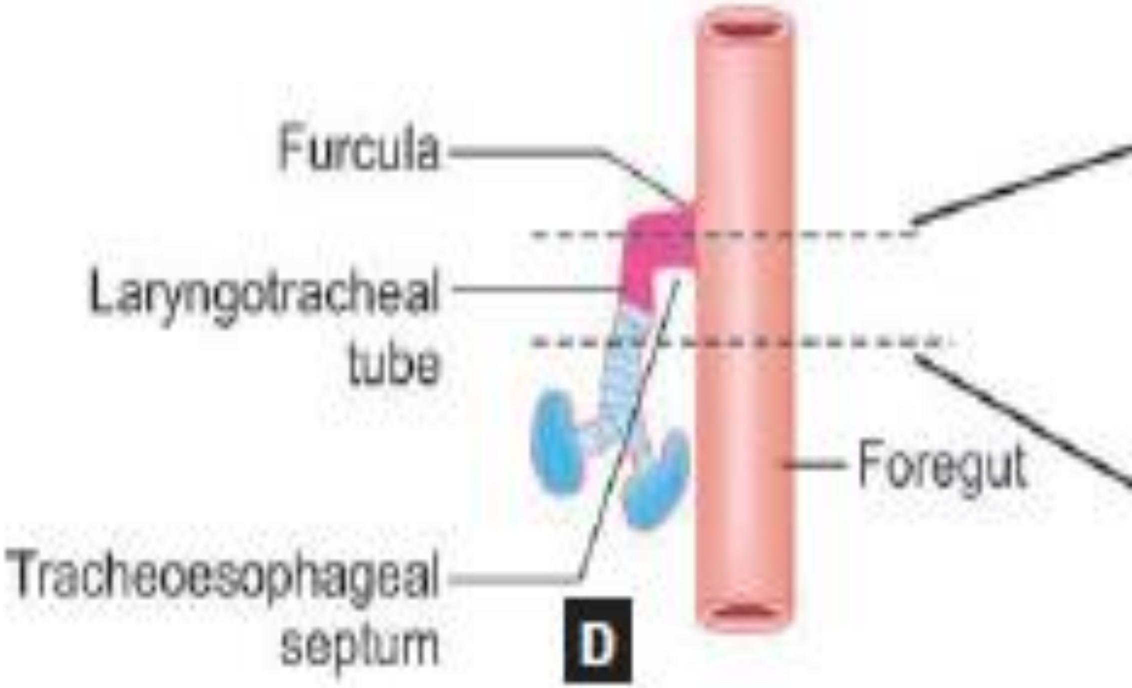
Fig. 16.2 Formation of laryngotracheal groove.



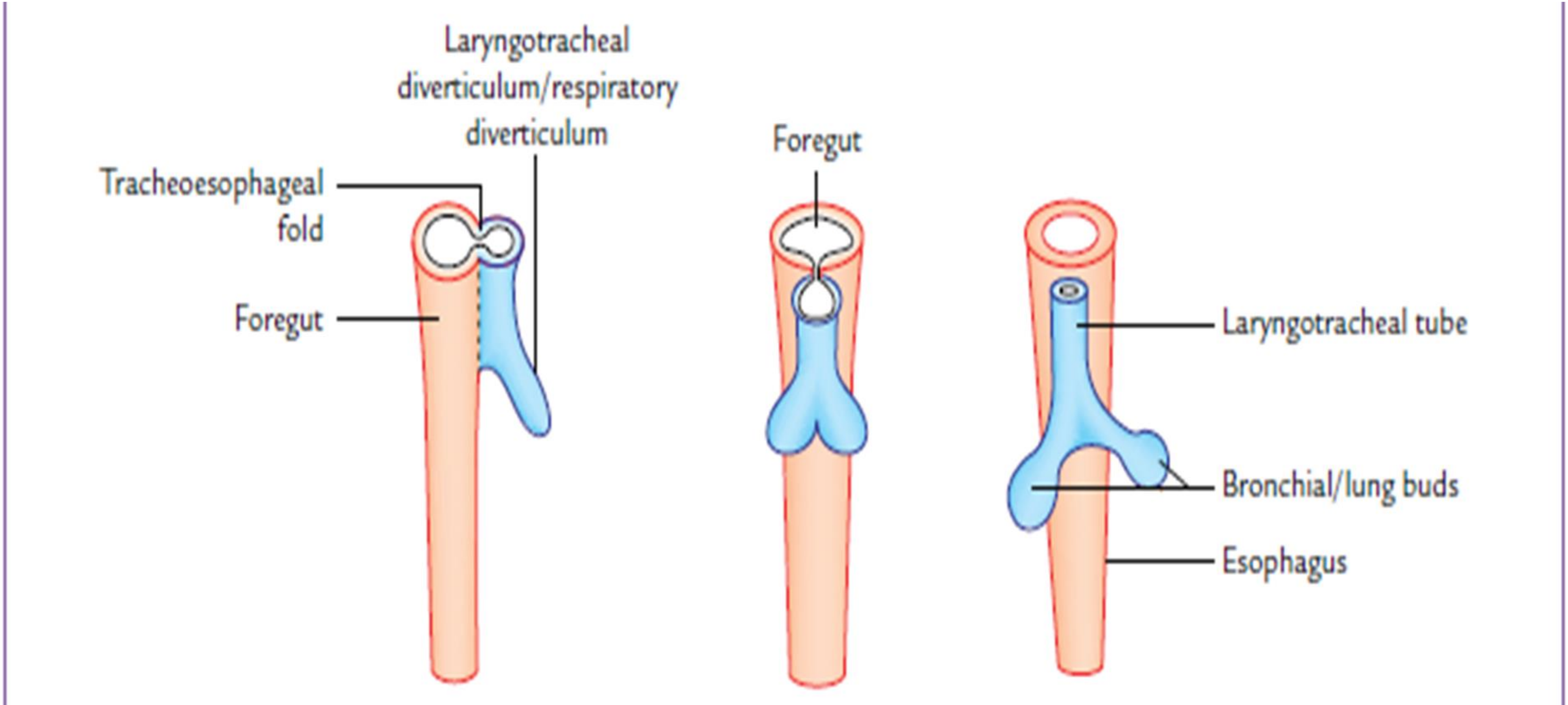
Laryngeal diverticulum
Tracheoesophageal fold
Tracheo-esophageal septum
Foregut
B
Laryngeal diverticulum
Laryngotracheal groove
Foregut
C
Furcula
Furcula
Laryngotracheal tube
Foregut
Tracheoesophageal septum
D

- ***Derivatives of respiratory diverticulum:***
- free caudal end grows downward to enter the thorax
- becomes bifid to form the right & left *bronchial/ lung buds*
- part of diverticulum cranial to the bifurcation is laryngotracheal tube & it forms the *larynx* & *trachea*
- *lung buds* form the *bronchi* & *lung parenchyma*





- groove separating ventral pharyngeal wall & proximal part of respiratory diverticulum is *laryngotracheal groove*
- laryngotracheal groove is bordered by 6th pharyngeal arch on either side



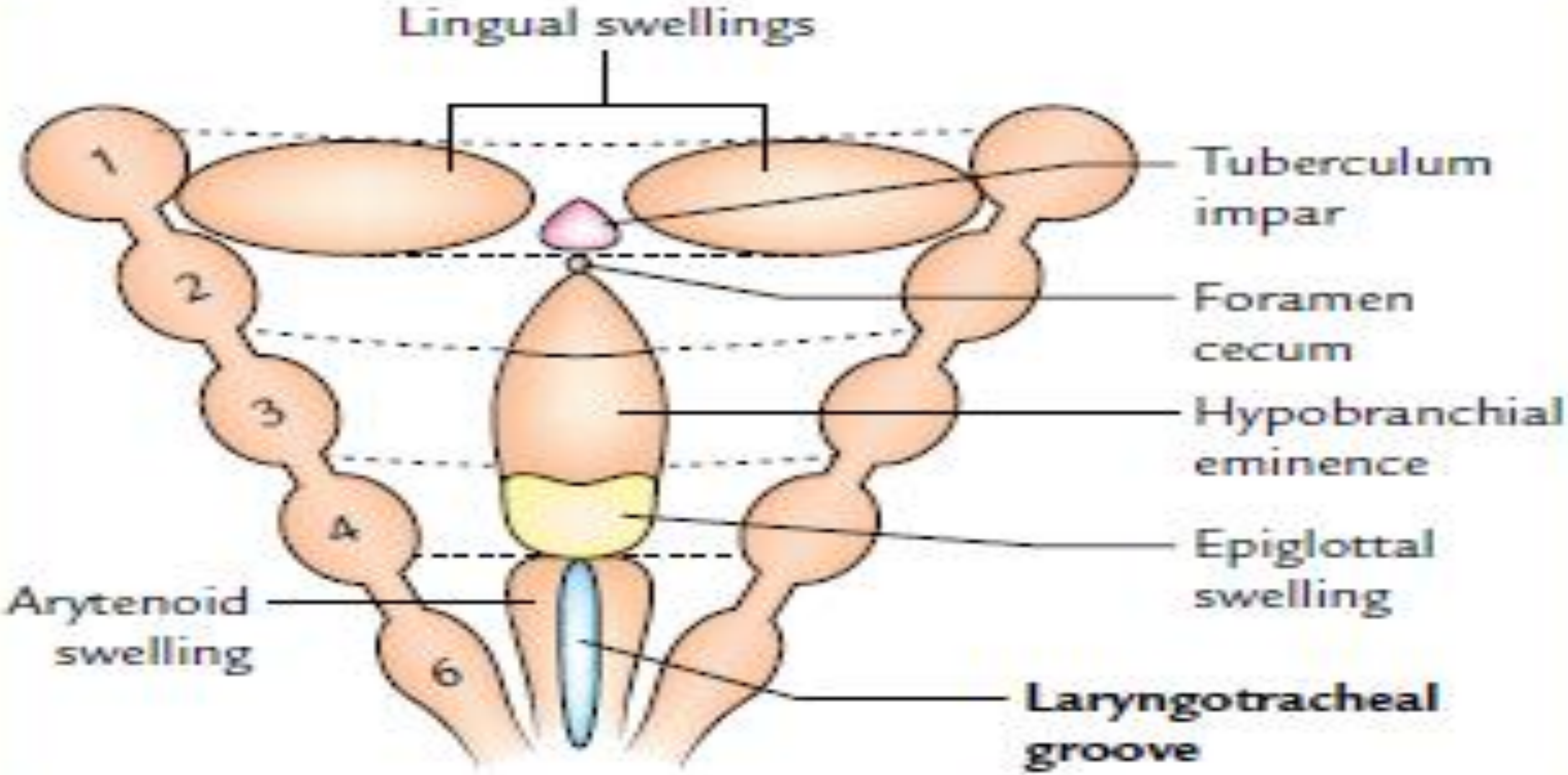
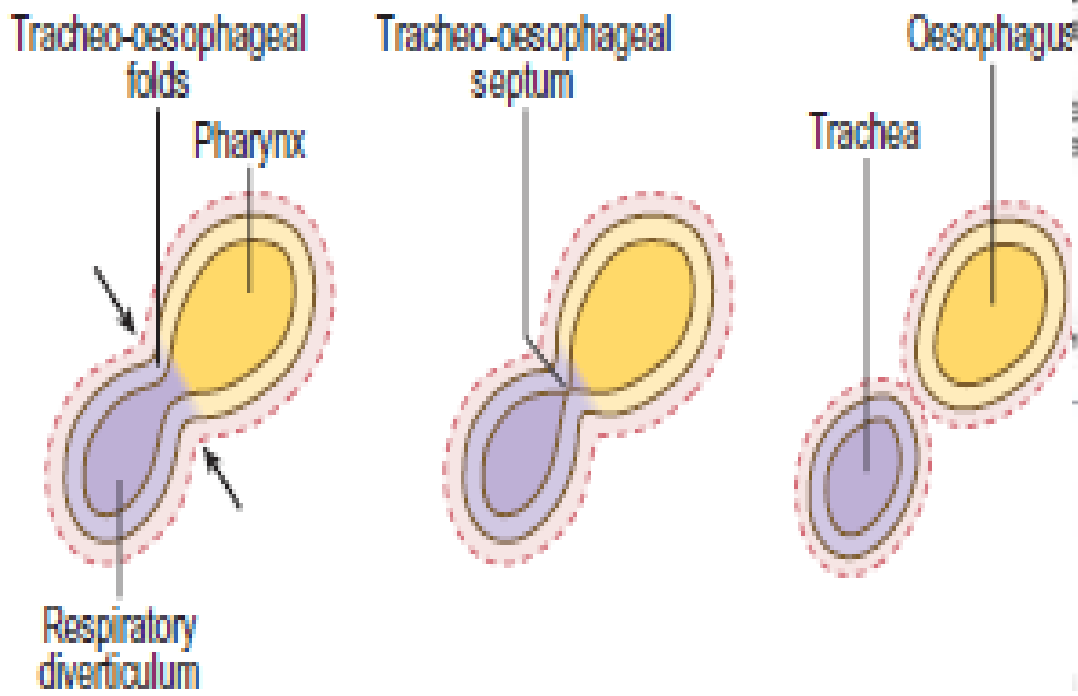
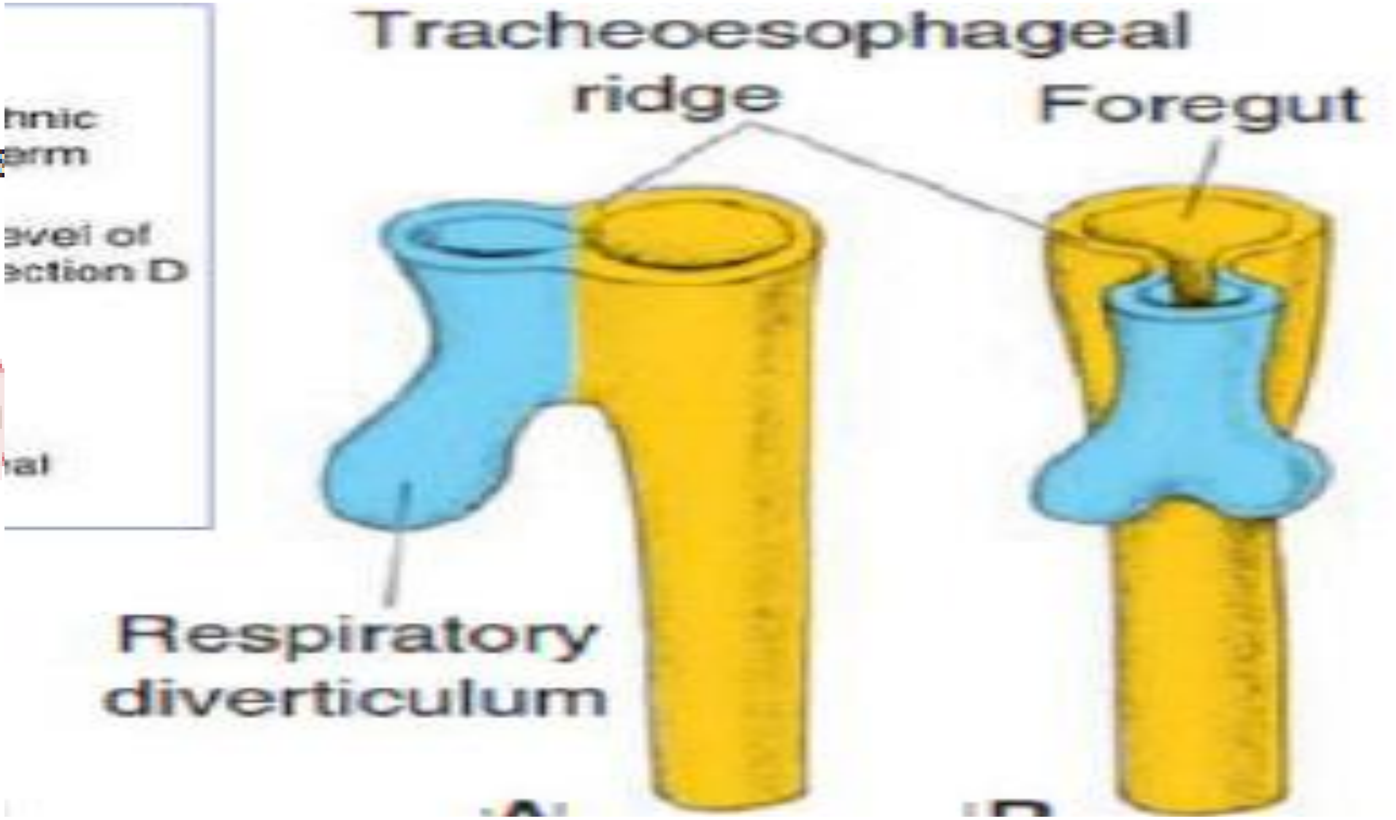


Fig. 16.2 Formation of laryngotracheal groove.

- caudal part of respiratory diverticulum 2 lateral folds- ***tracheooesophageal folds*** grow medially from the margins of the groove
- fuse to form ***tracheoesophageal septum***
- tracheoesophageal septum lies in the coronal plane & extends caudocranially
- caudal part of septum separates trachea ventrally from the esophagus dorsally
- opening of the diverticulum into the primitive pharynx becomes the **laryngeal orifice or inlet of larynx**





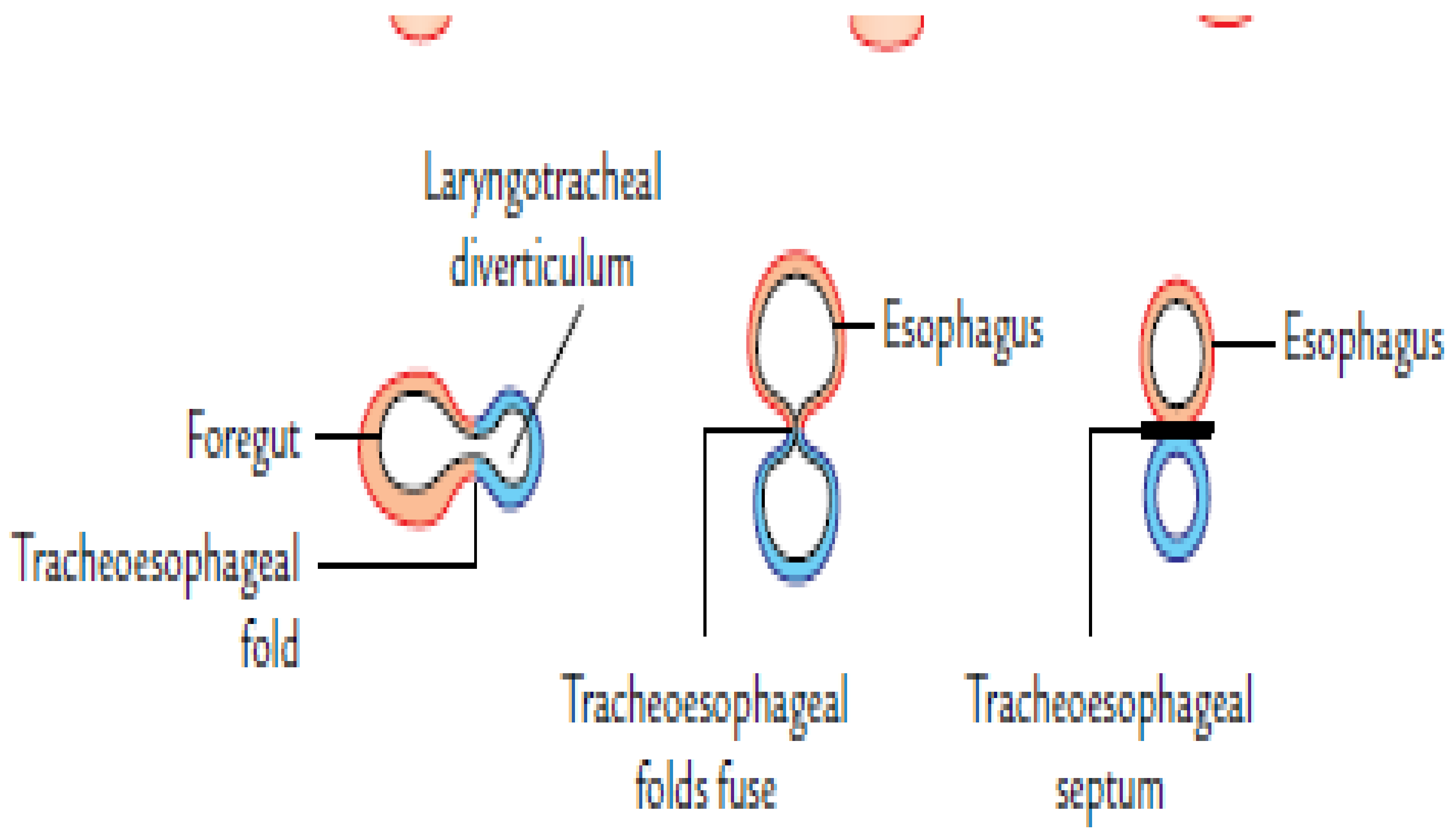
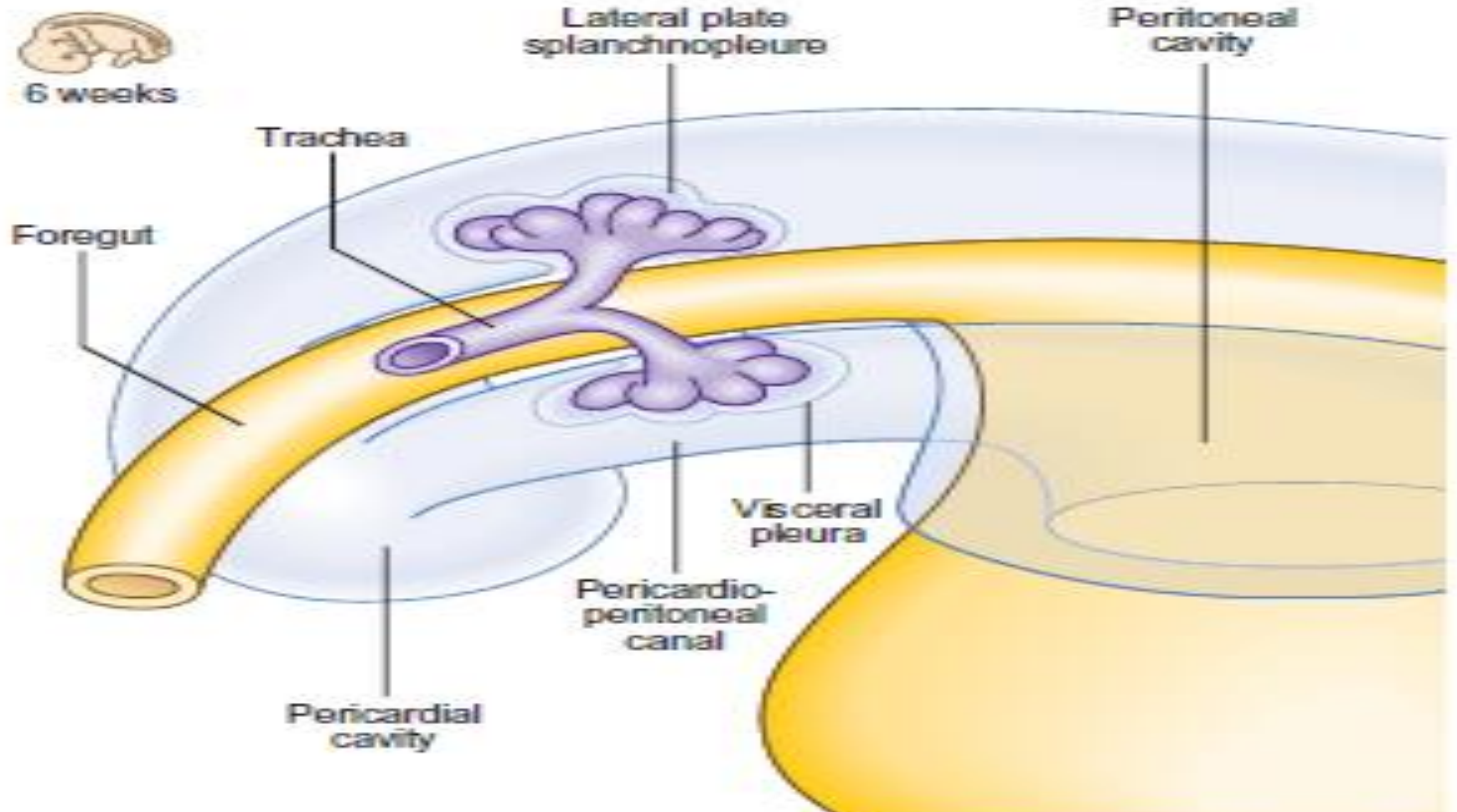
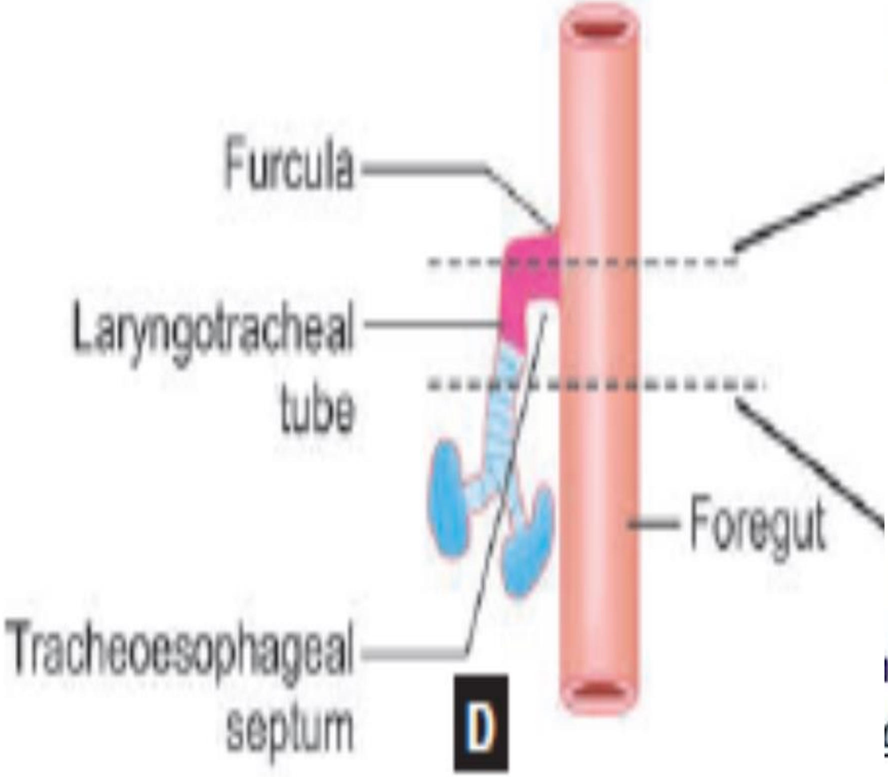


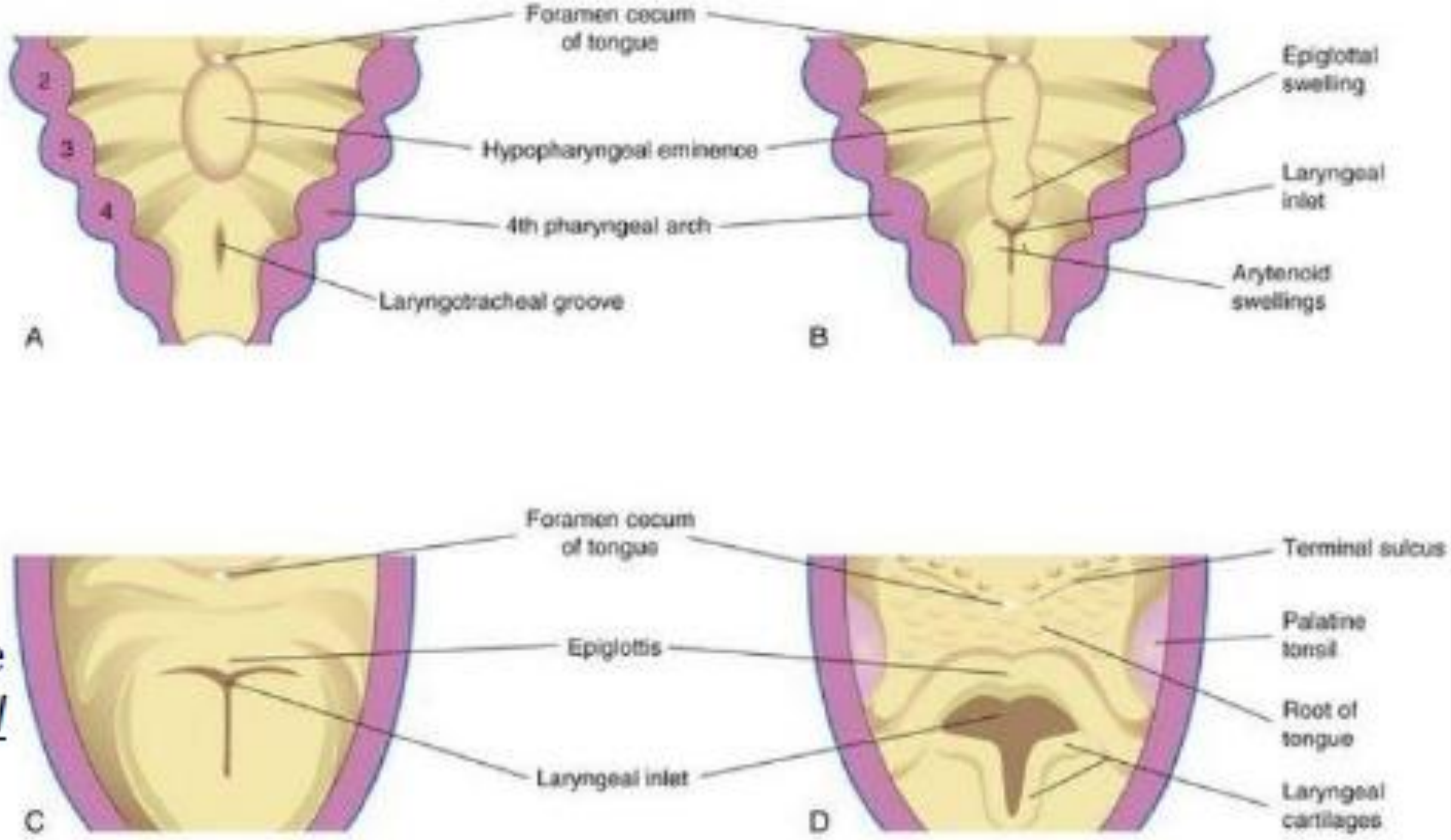
Fig. 16.3 Successive stages in the development of respiratory diverticulum. Figures below are transverse section of the above figures to show the formation of tracheoesophageal septum and appreciate how it separates foregut into trachea and esophagus.

- cranial part of tracheoesophageal septum is arrested to form a sagittal slit the ***inlet of larynx*** **or** ***furcula of His***
- through which laryngotracheal tube communicates with the pharynx
- Each lung bud invaginates into the pericardioperitoneal canal
- right & left pericardioperitoneal canals form right & left *pleural cavities*



- **LARYNX**
- develops from cranial most part of the laryngotracheal tube
- *Inlet of larynx*: surrounded by the mesenchyme of 4th & 6th pharyngeal arches derived from neural crest cells
- Mesenchyme proliferates to form laryngeal cartilages





- slit-like **laryngeal inlet** becomes *U*-shaped
- bounded at its cephalic end by caudal part of hypobranchial eminence (4^{th} arch derivative)
- on each side by 6th arch derived, arytenoid swellings
- proliferation of mesenchyme of 4th & 6th arches makes the *U*-shaped laryngeal inlet into a *T*-shaped orifice
- horizontal limb of T is caudal to hypobranchial eminence & the vertical limb between arytenoid swellings

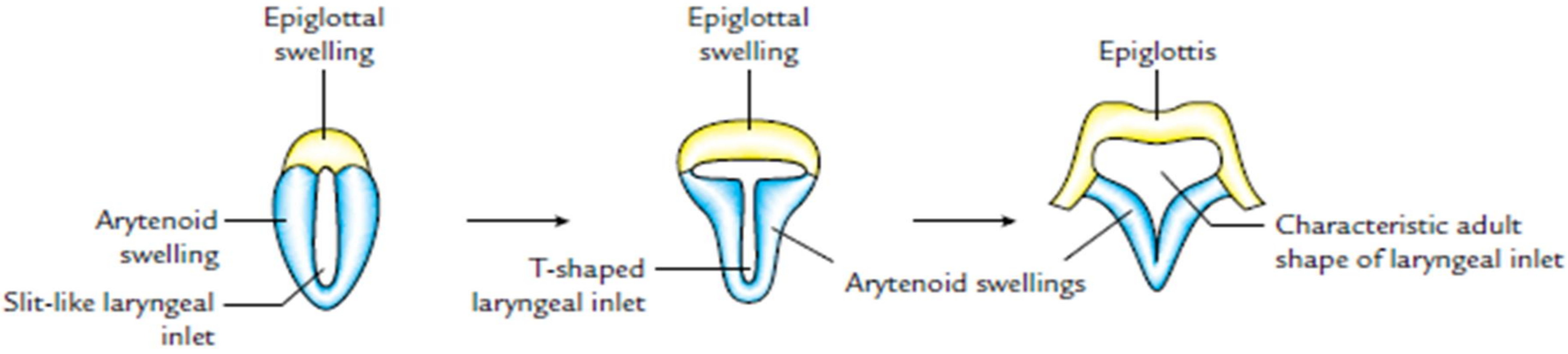
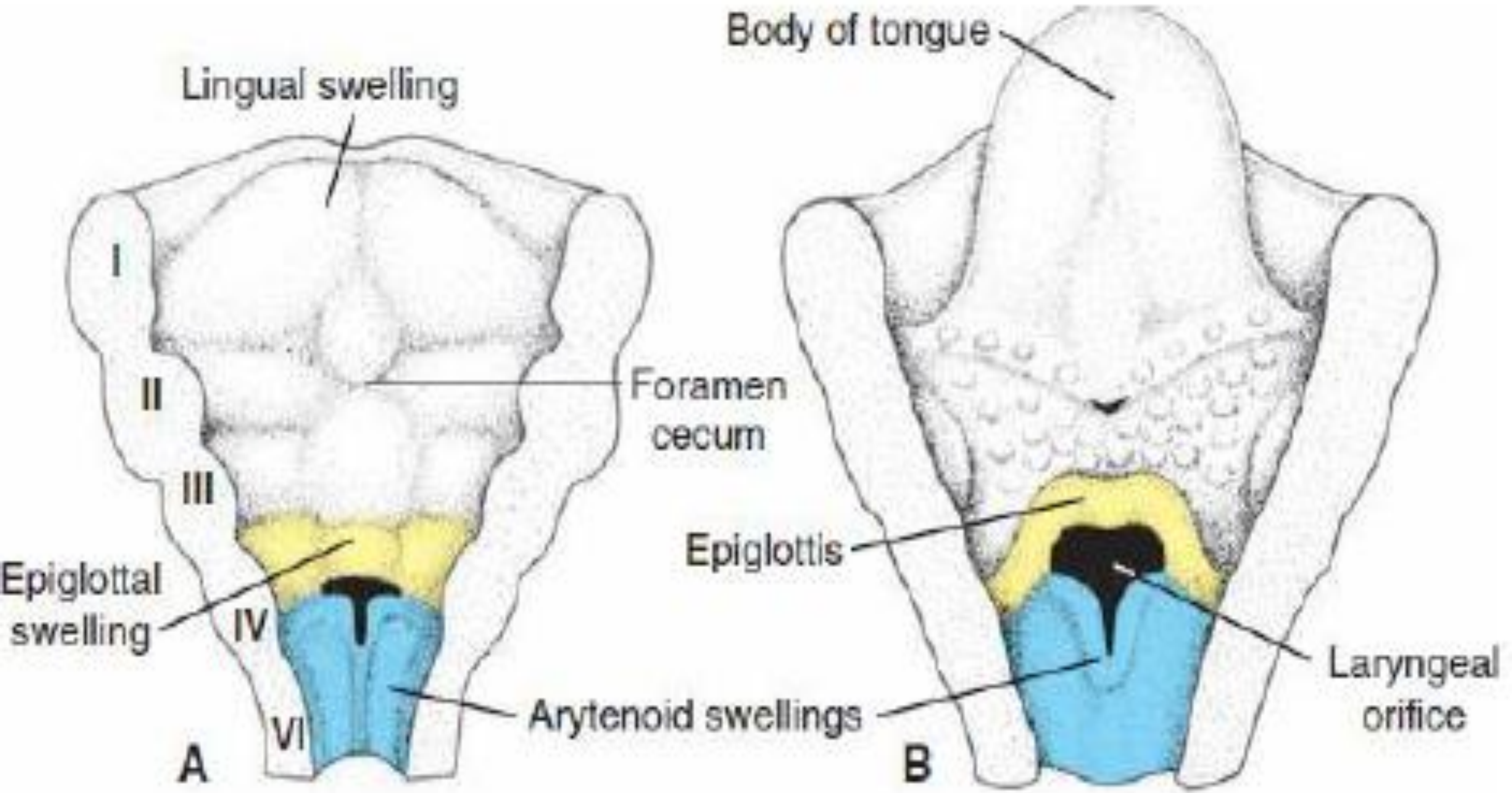
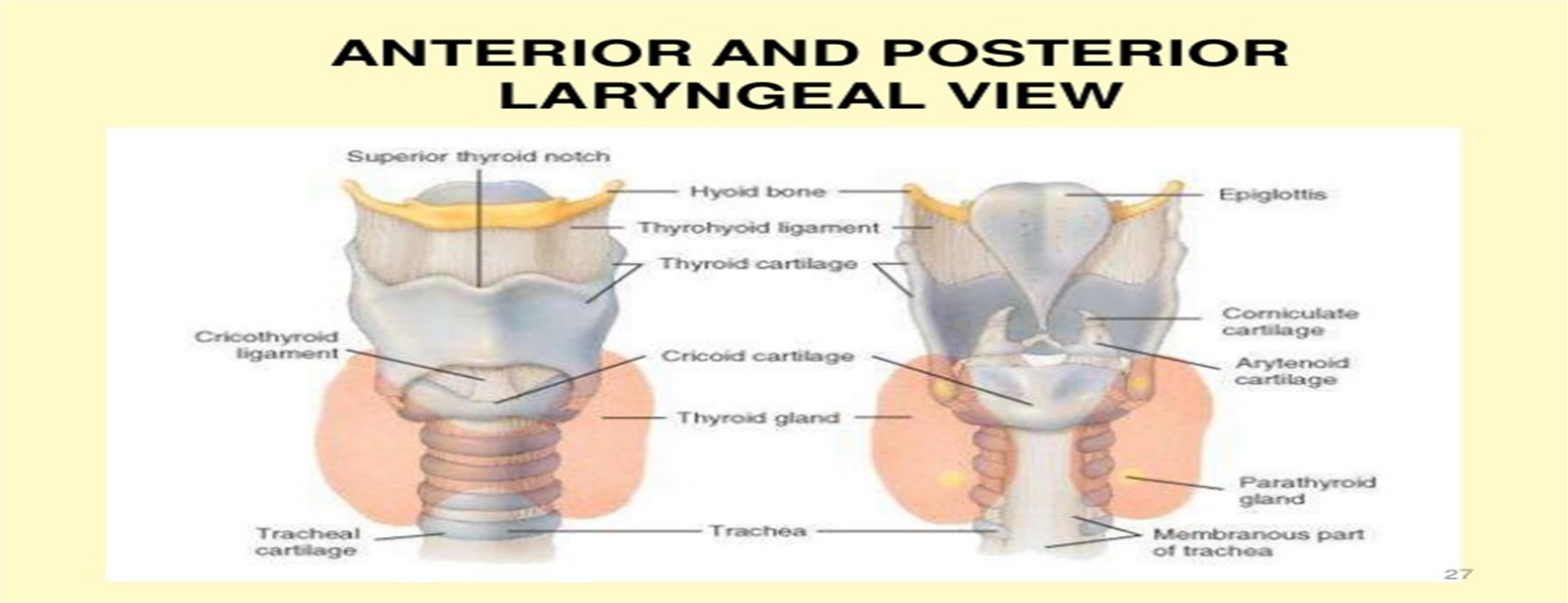


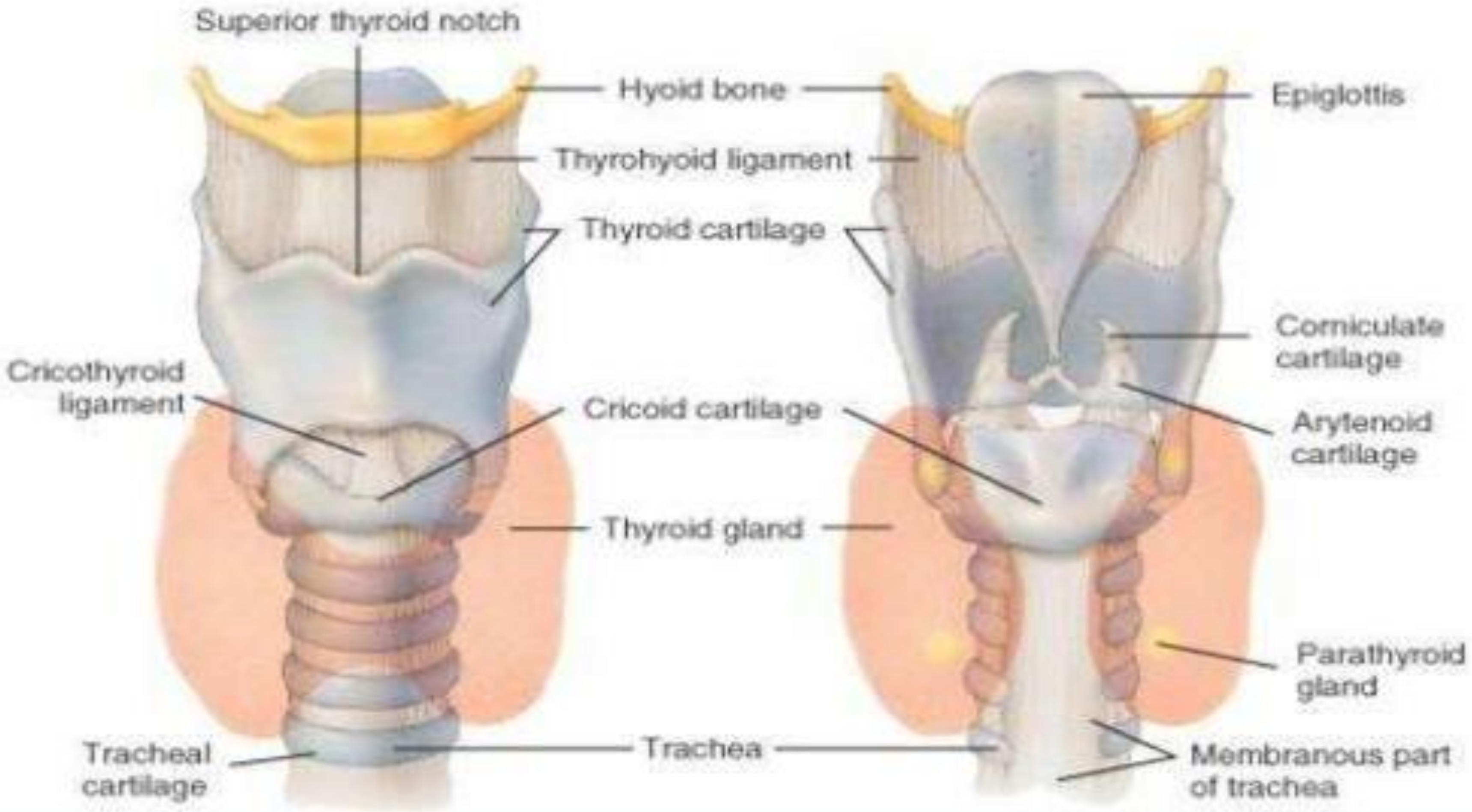
Fig. 16.4 Change in shape of laryngeal inlet during development of larynx.


Body of tongue
Lingual swelling
I
II
III
IV
VI
Foramen
cecum
Epiglottal
swelling
Arytenoid swellings
Epiglottis
Laryngeal
orifice
A
B

- Reorganization of cartilages results in characteristic adult shape of laryngeal inlet
- caudal part of hypobranchial eminence (4th arch) forms epiglottis, thyroid & cuneiform cartilages
- upper part of arytenoid swelling forms the arytenoid & corniculate cartilages, lower part forms the cricoid cartilage derivatives of *6th arch*
- *Lining epithelium*: derived from *endoderm*



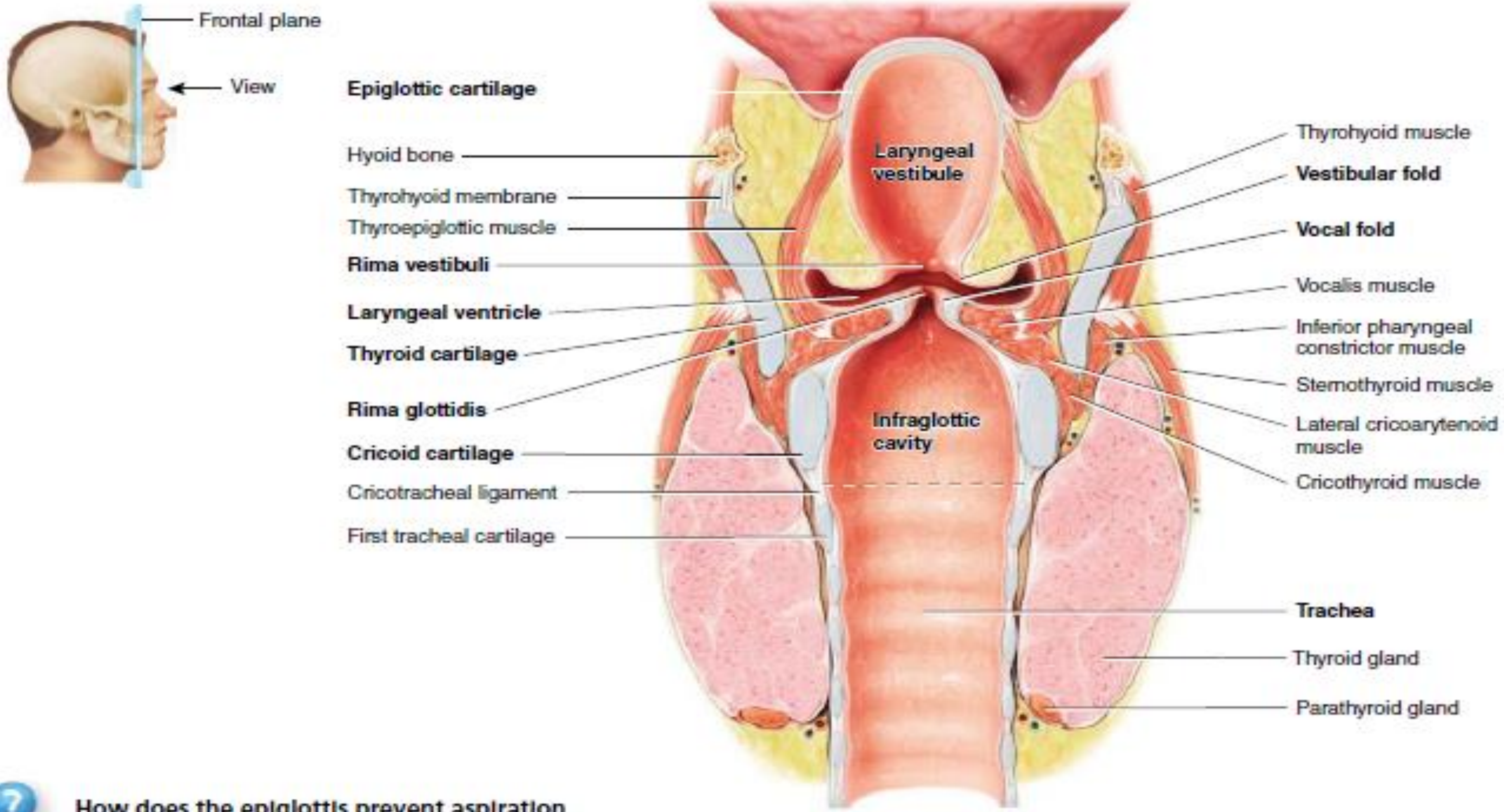
ANTERIOR AND POSTERIOR LARYNGEAL VIEW



- *Muscles* are derived from branchial mesoderm as indicated by their nerve supply
- (All intrinsic muscles supplied by *recurrent laryngeal nerve* (6th arch) except cricothyroid which is supplied by external laryngeal branch of *superior laryngeal nerve* (4th arch)

Intrinsic muscles of the larynx				
Muscle	**Origin**	**Insertion**	**Innervation**	**Function**
Cricothyroid	Anterolateral aspect of arch of cricoid cartilage	Oblique part-lesser horn of hyoid; straight part-inferior margin of thyroid cartilage	External branch of superior laryngeal nerve from the vagus nerve [X]	Forward and downward rotation of the thyroid cartilage at the cricothyroid joint
Posterior cricoarytenoid	Oval depression on posterior surface of lamina of cricoid cartilage	Posterior surface of muscular process of arytenoid cartilage	Recurrent laryngeal branch of the vagus nerve [X]	External rotation and abduction of the arytenoid cartilage
Lateral cricoarytenoid	Superior surface of arch of cricoid cartilage	Anterior surface of muscular process of arytenoid cartilage	Recurrent laryngeal branch of the vagus nerve [X]	Abduction and internal rotation of the arytenoid cartilage
Transverse arytenoid	Lateral border of posterior surface of arytenoid cartilage	Lateral border of posterior surface of opposite arytenoid cartilage	Recurrent laryngeal branch of the vagus nerve [X]	Adduction of arytenoid cartilages
Oblique arytenoid	Posterior surface of muscular process of arytenoid cartilage	Posterior surface of apex of adjacent arytenoid cartilage; extends into aryepiglottic fold	Recurrent laryngeal branch of the vagus nerve [X]	Sphincter of the laryngeal inlet
Thyro-arytenoid	Thyroid angle and adjacent cricothyroid ligament	Anterolateral surface of arytenoid cartilage; some fibers continue in aryepiglottic folds to the lateral margin of the epiglottis	Recurrent laryngeal branch of the vagus nerve [X]	Sphincter of vestibule and of laryngeal inlet
Vocalis	Lateral surface of vocal	Vocal ligament and thyroid	Recurrent laryngeal branch of the	Adjusts tension in vocal folds

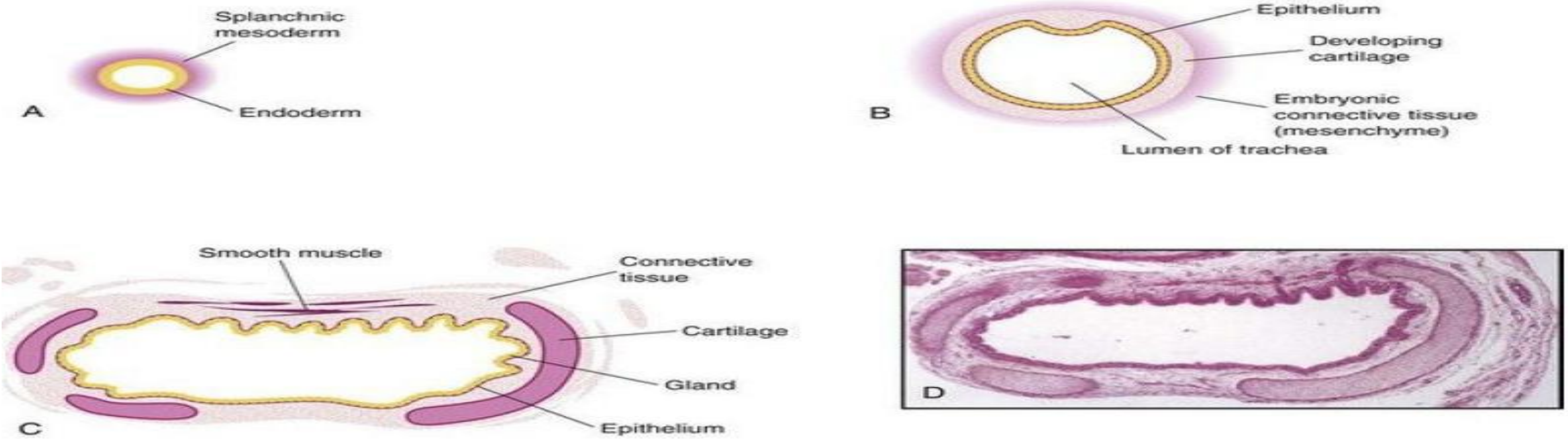
- ***Interior of larynx*:**
- proliferation of epithelium of larynx occludes its lumen
- recanalization occurs 10th week produces 2 lateral recesses - ***ventricles of larynx***
- recesses are bounded by endodermal folds that differentiate to form an upper pair of **vestibular folds** & a lower pair of **vocal folds**
- vocal folds are at the junction of 4th & 6th arches
- sensory innervation of mucosa of larynx:-
- above the vocal folds is from **internal laryngeal branch of vagus** (4th arch)
- below the vocal folds is from **recurrent laryngeal branch of vagus** (6th arch)



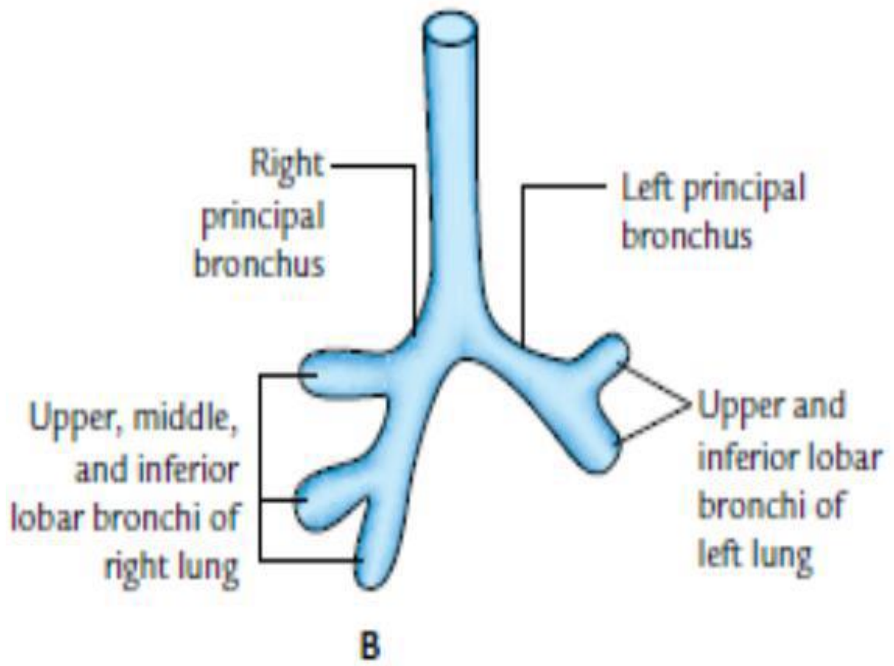
(d) Frontal section

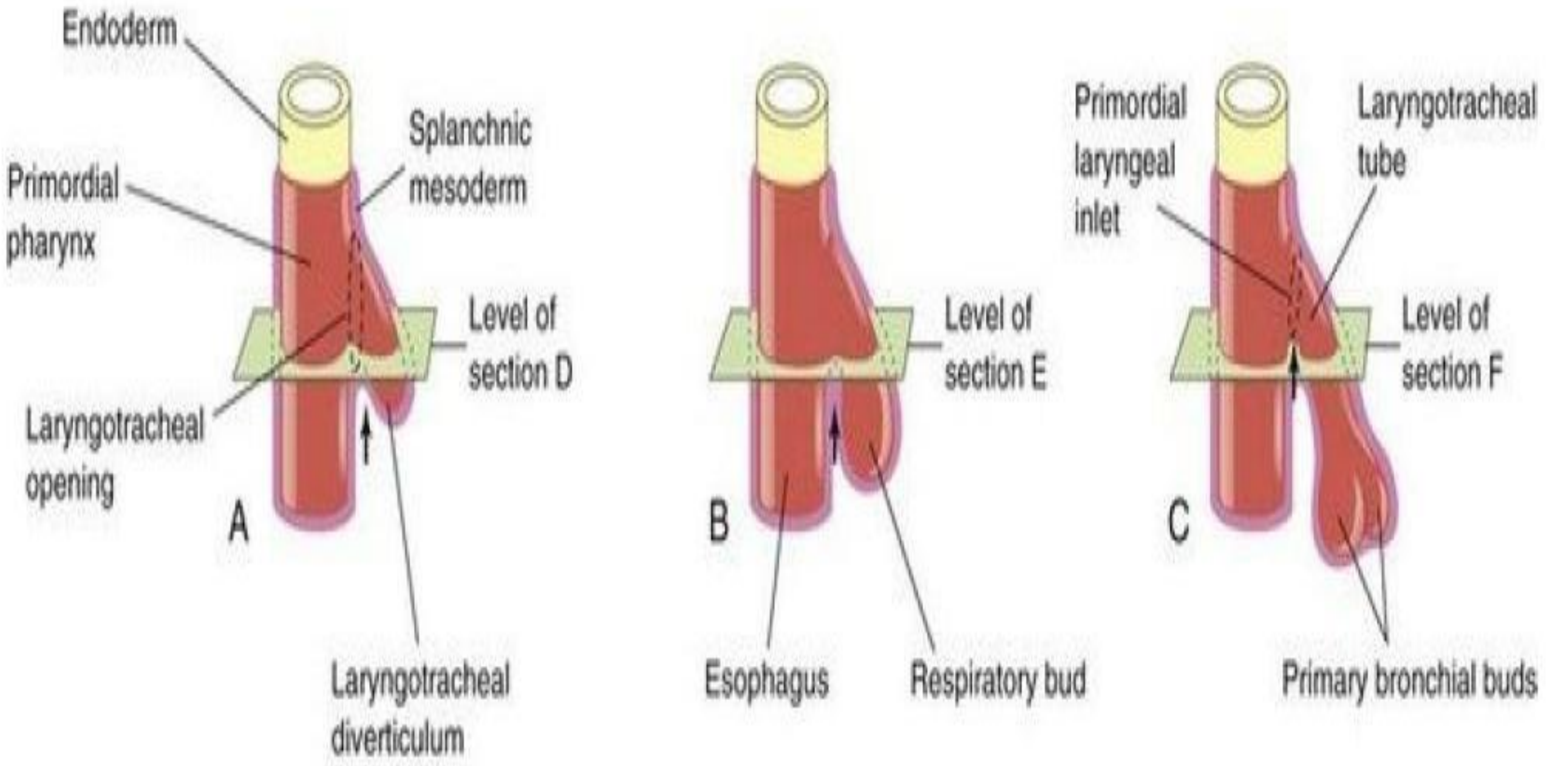
How does the epiglottis prevent aspiration of foods and liquids?

- **TRACHEA**
- develops from part that lies between *bronchial or lung buds* & larynx
- elongates due to caudocranial extension of tracheoesophageal septum
- At birth bifurcation of trachea lies at the level of lower border of T4
- endoderm of laryngotracheal diverticulum forms the lining epithelium & glands of trachea
- cartilage, muscle & connective tissue develop from splanchnopleuric mesoderm surrounding laryngotracheal tube

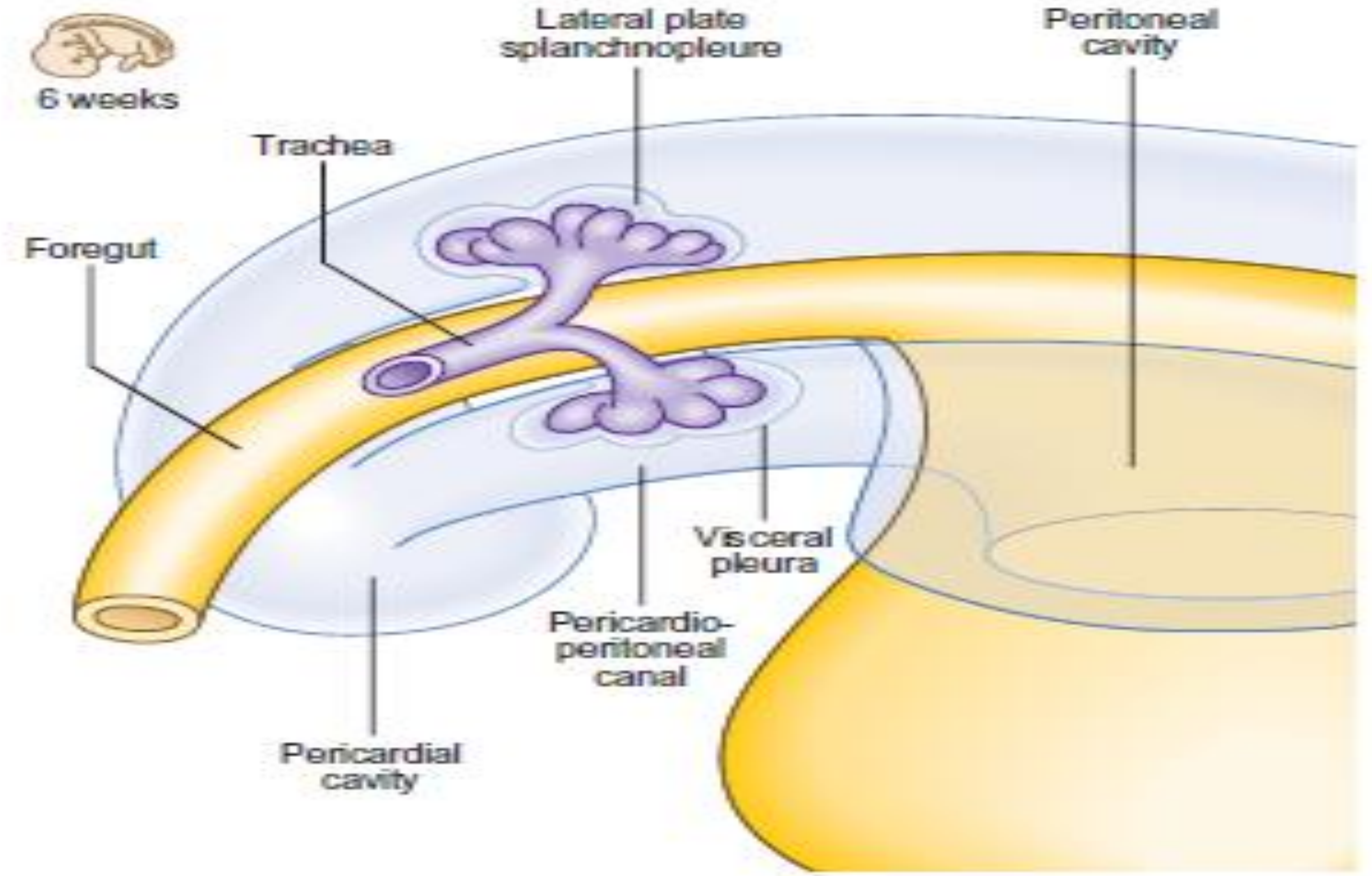


- **EXTRAPULMONARY BRONCHI**
- laryngotracheal diverticulum divides during 5th week form the *right & left principal bronchi*
- right principal bronchus is larger than the left & is in line with the trachea
- left division comes to lie more transversely

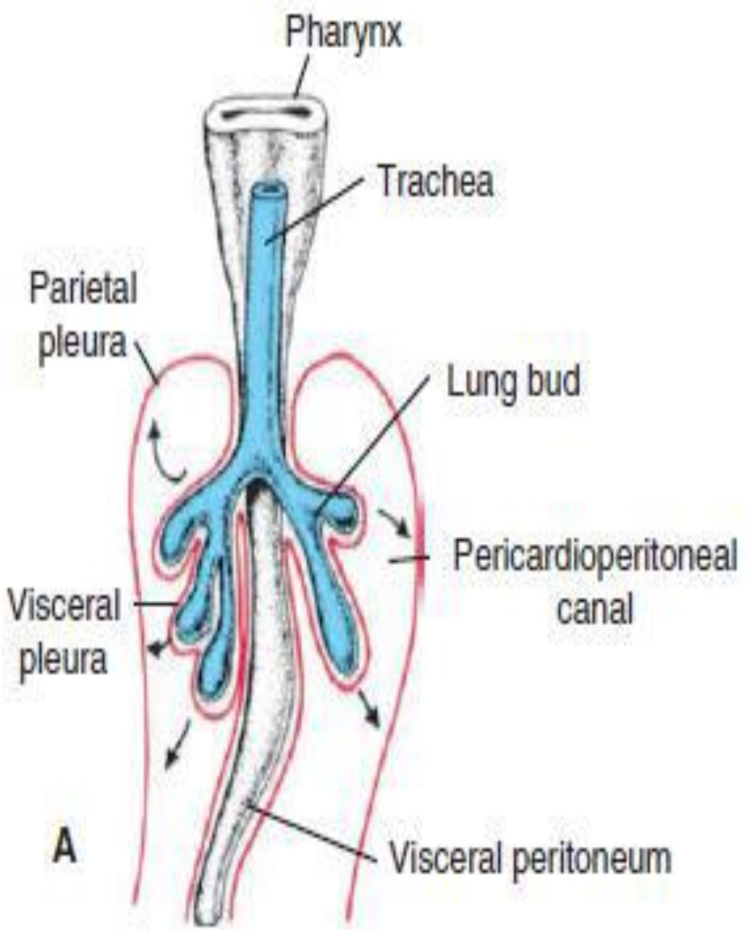


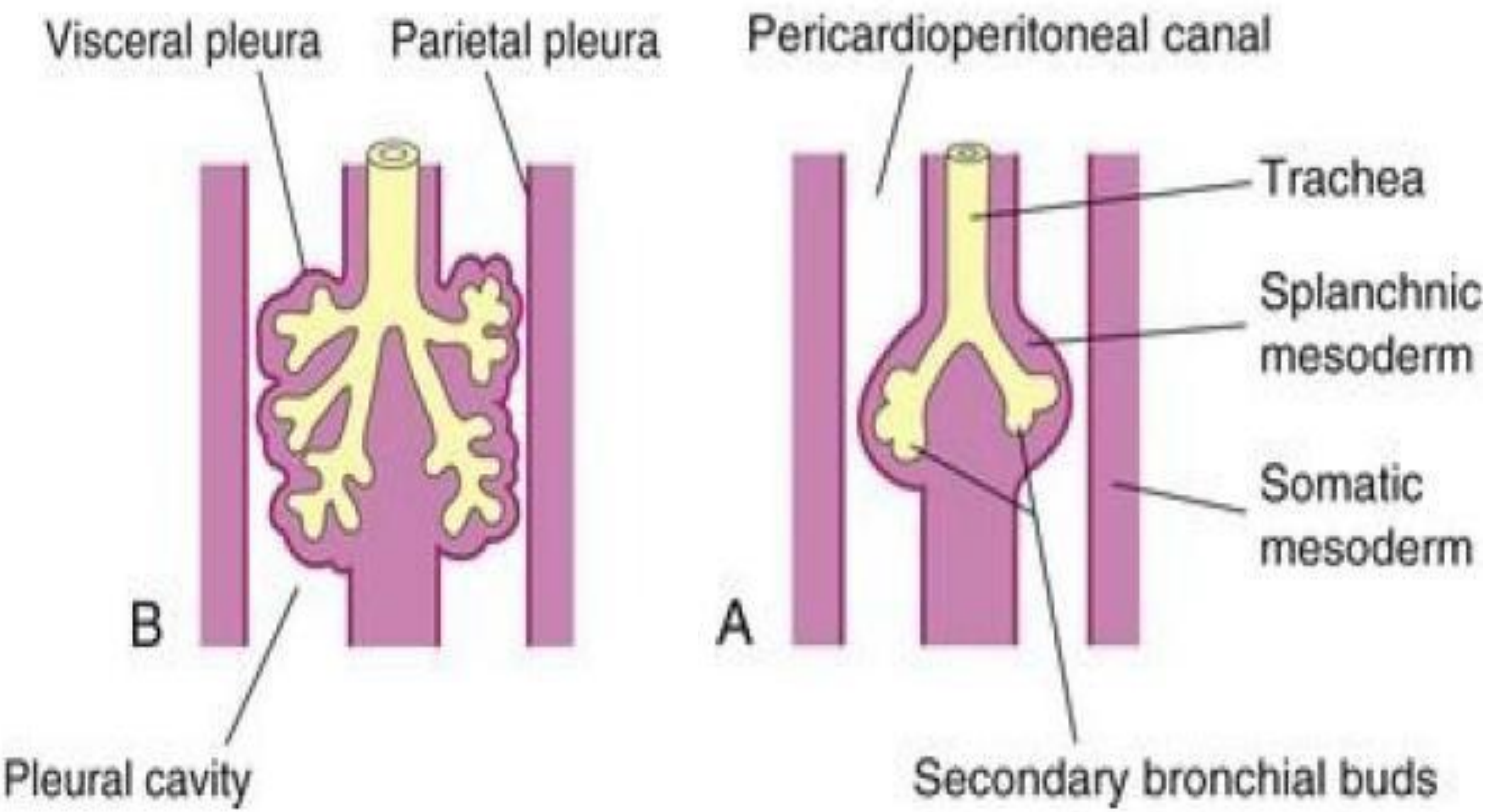


- **INTRAPULMONARY BRONCHI AND LUNGS**
- ***Formation of pleural cavity and pleura:***
- respiratory diverticulum grows in caudal & lateral direction
- lung buds come in contact with the respective ***pericardioperitoneal canal*** *(primordial of pleural cavities)*& bulges into it
- lie on either side of the foregut

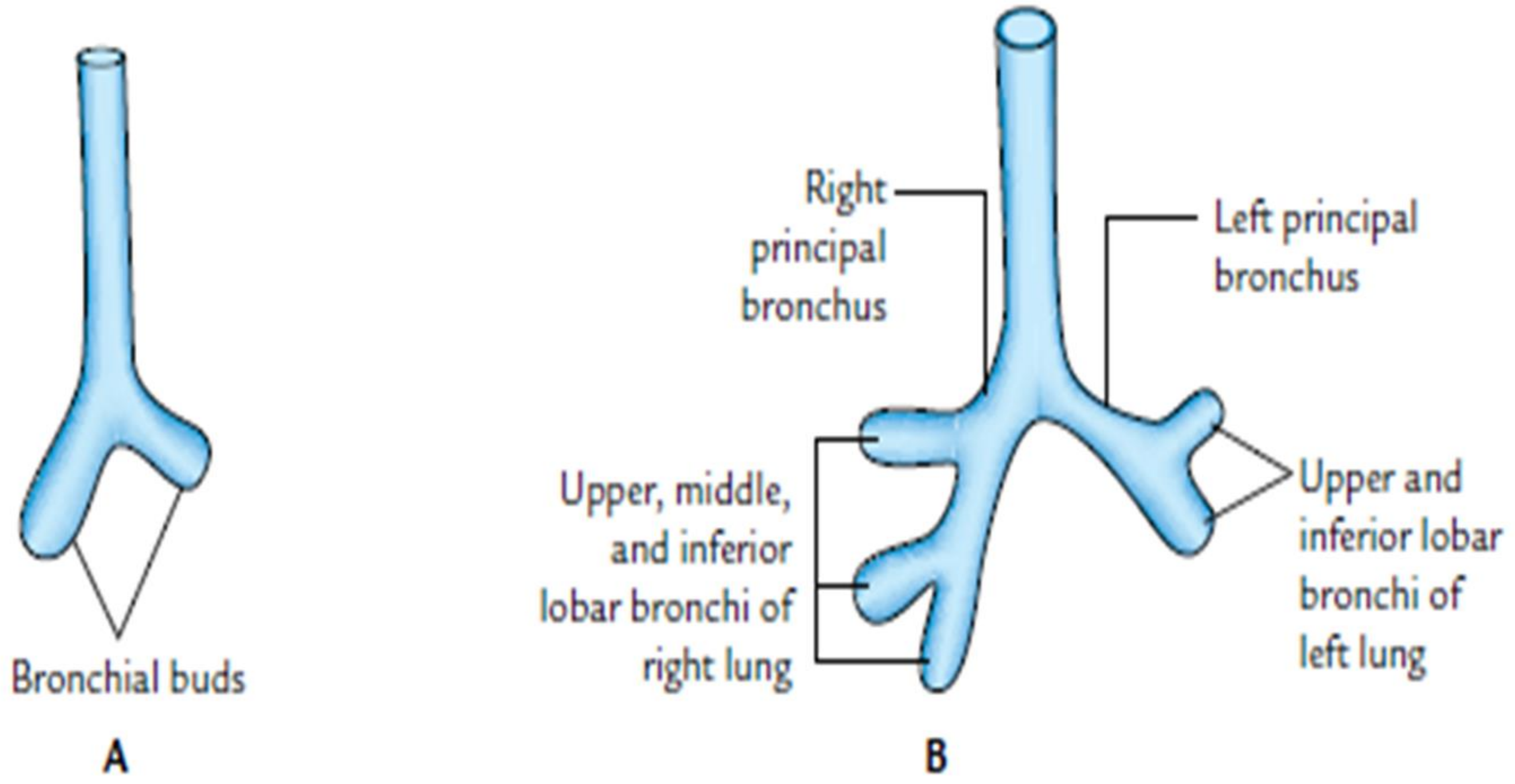


- **splanchnopleuric layer** of intraembryonic mesoderm in contact with the dividing & expanding lung bud becomes the *visceral pleura*
- **somatopleuric layer** of intraembryonic mesoderm covering the inner aspect of body wall forms the *parietal pleura*
- space between the parietal & visceral pleura forms the ***pleural cavity(space)***



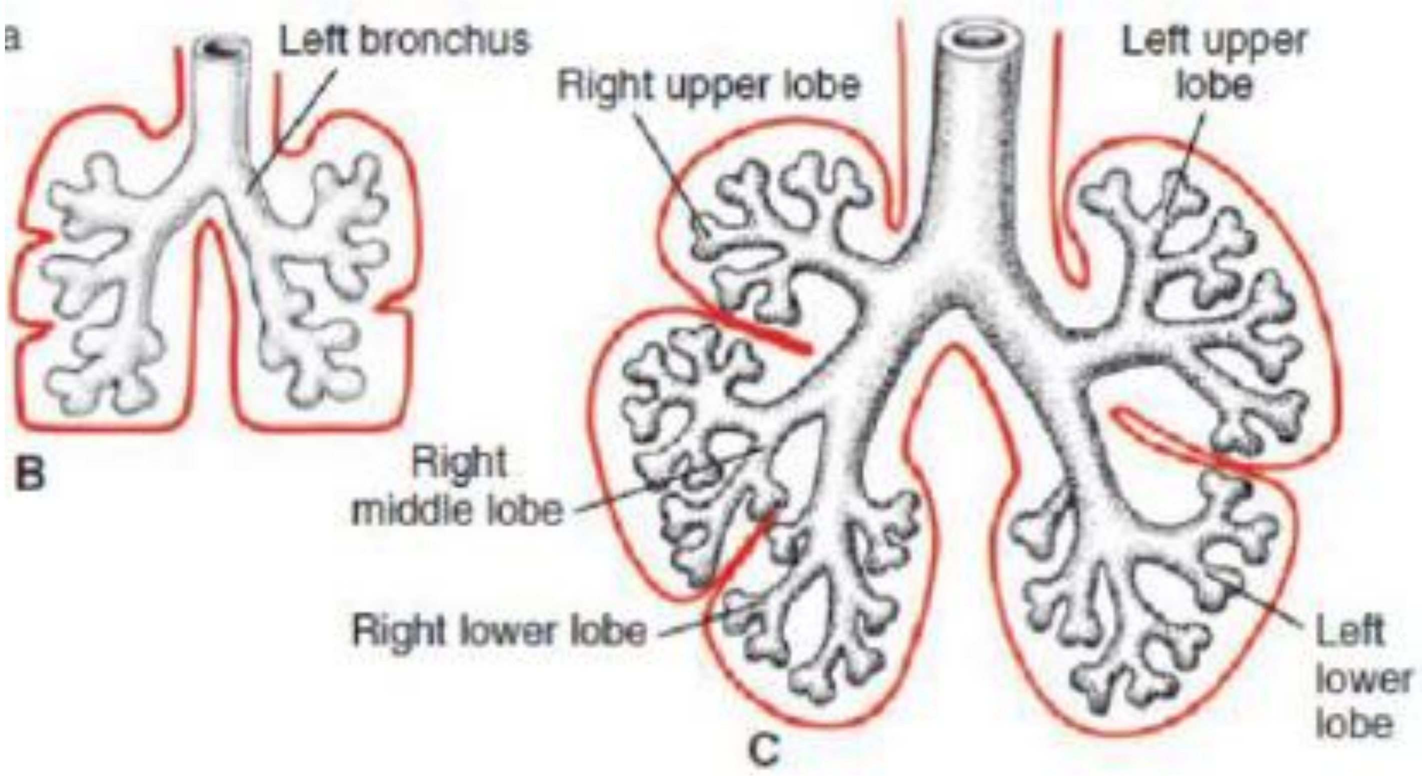


- ***Subdivisions of intrapulmonary bronchi***:
- right & left principal bronchi subdivide into secondary/*lobar bronchi*
- left principal bronchus shows 2 subdivisions (upper & lower) that represent the 2 lobar bronchi of the left lung
- right division divides into 3 lobar bronchi (superior, middle & inferior) of right lung

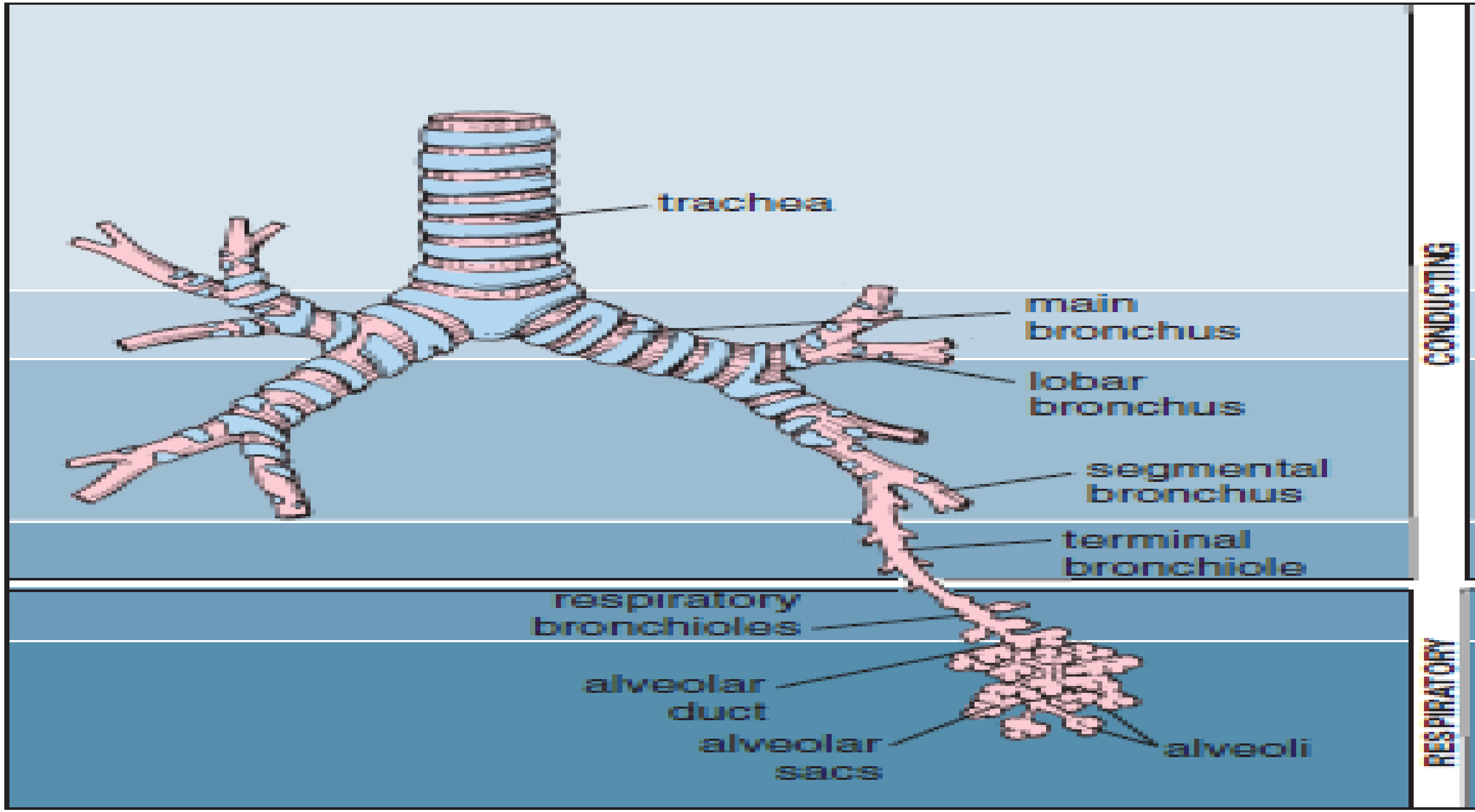


- pleura lines the surface of each lobe , lobes come to be separated by fissures
- substance of the lung is formed by further subdivisions of the lobar bronchi
- Each lobar bronchus subdivides to form *segmental bronchi* in the 7th week
- segmental bronchus & its surrounding mesenchyme constitute the *bronchopulmonary segment*
- Each lung contains 10 bronchopulmonary segments

RIGHT LUNG		LEFT LUNG	
Lobe	Segments	Lobe	Segments
Superior	Apical	Superior	Apical
	Posterior		Posterior
	Anterior		Anterior
Middle	Lateral		Sup. lingular
	Medial		Inf. lingular
Inferior	Superior	Inferior	Superior
	Medial basal		Medial basal
	Anterior basal		Anterior basal
	Lateral basal		Lateral basal
	Posterior basal		Posterior



- total number of divisions of each main bronchus is about 17 by 7th month
- 6 more after birth to attain final shape of adult lung
- Divisions of each segmental bronchus form the *bronchioles*, *terminal bronchioles, respiratory bronchioles, alveolar ducts and alveolar sacs & alveoli*
- After establishment of the bronchial tree, alveoli are formed by expansion of the terminal parts of the tree



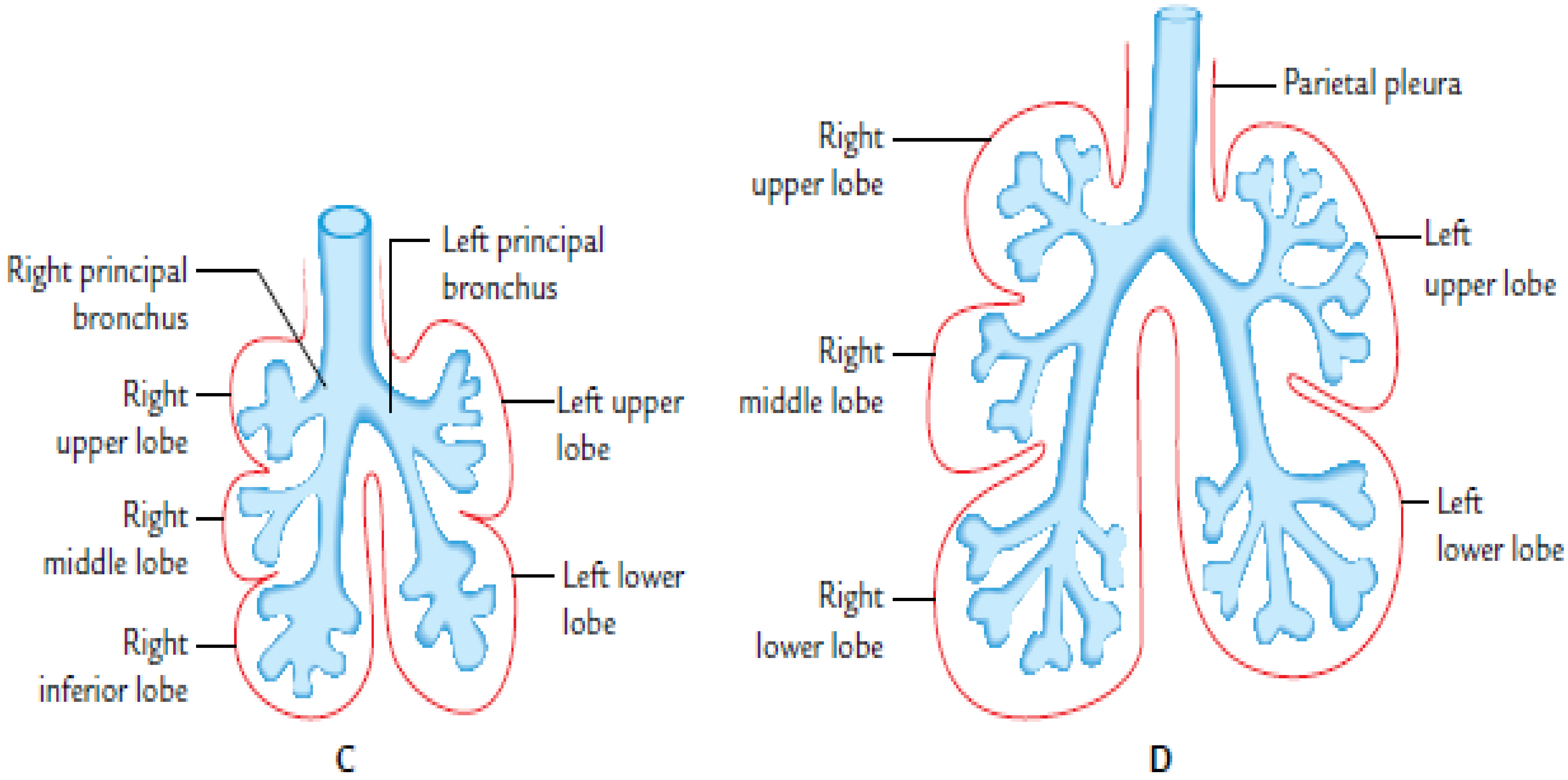
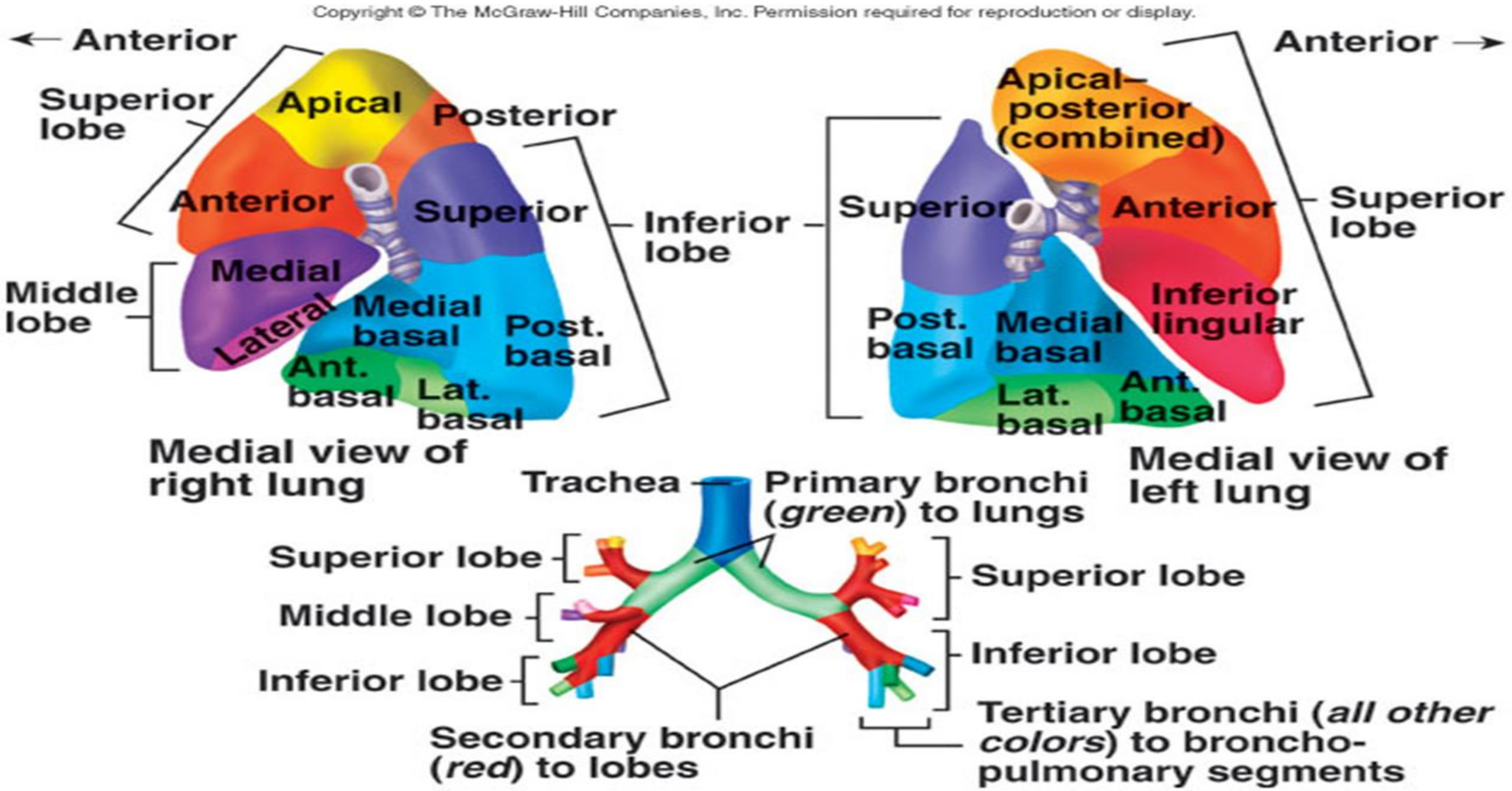


Fig. 16.7 Successive stages in the development of bronchi and lungs. A. At 4 weeks. B. At 5 weeks. C. At 6 weeks. D. At 8 weeks.

- ***Parenchyma of lung***:
- parts of the lung parenchyma, developing from the endoderm of lobar bronchi, are separated from one another by mesoderm
- endoderm of respiratory diverticulum forms the lining epithelium of the bronchial tree
- mesoderm forms cartilages, smooth muscle & connective tissue, basis of lung & also gives rise to the pleura

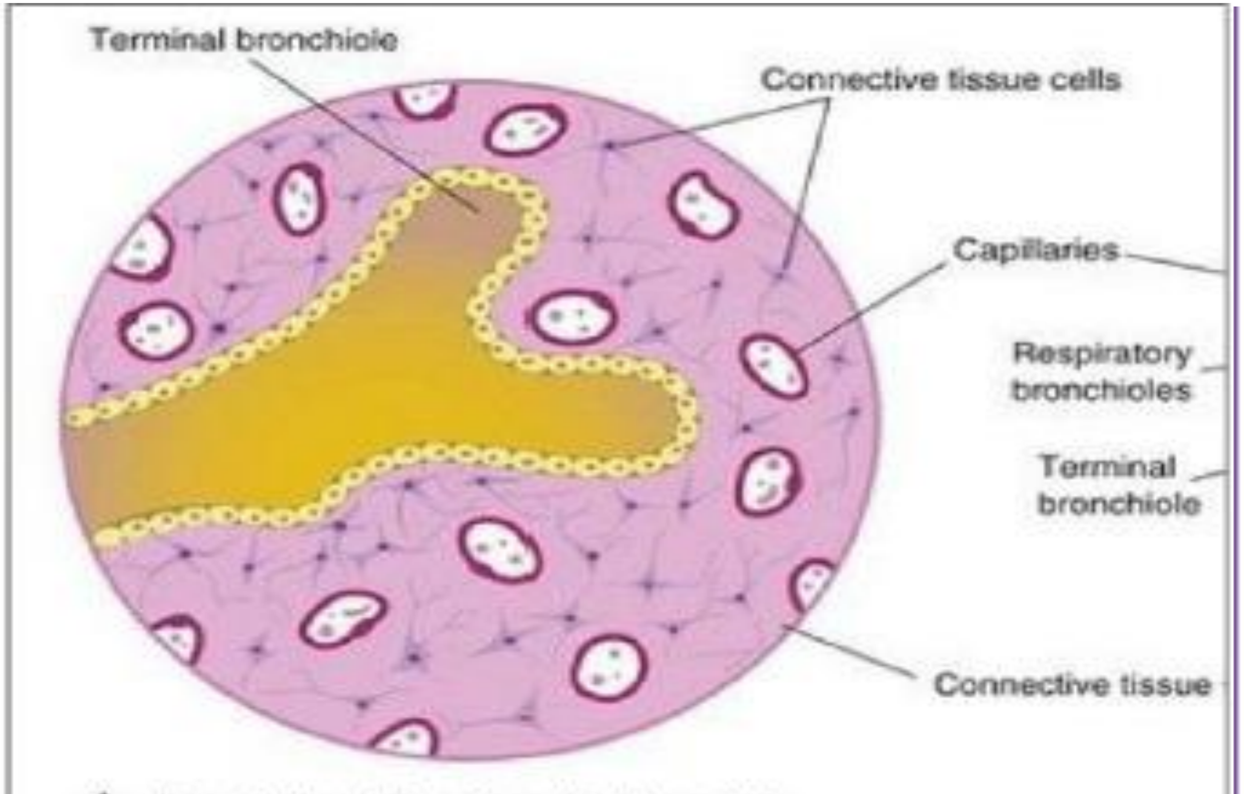


- ***Maturation of lung*****:**
- 4 stages
- ramifications of the bronchial tree pass through these stages
- fetal life all subdivisions of the bronchial tree are lined by cuboidal epithelium that changes with the maturation of lung
- some of the cells become specialized to for production of surfactant
- substance is rich in phospholipids & forms a thin layer over alveoli and reduces surface tension

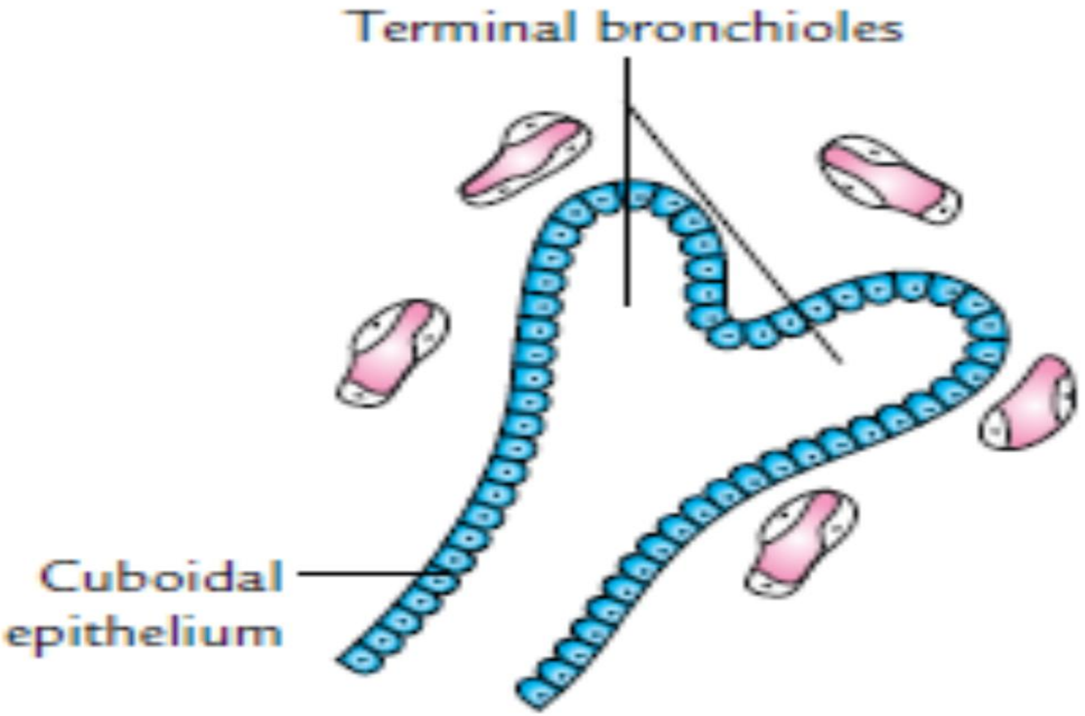
- ***Pulmonary surfactant***:
- Before birth respiratory passages are full of fluid derived from amniotic fluid which also contains surfactant
- surfactant remains as a thin layer lining the alveoli after birth
- prevents collapse of alveoli during expiration
- ***Viable age of fetus*:**
- pulmonary circulation is established early in fetal life

- Maturation of the lungs is divided into 4 stages/periods:
- pseudoglandular stage,
- canalicular stage,
- terminal sac stage
- alveolar stage

- **1. Pseudoglandular stage/period (5–16 weeks):**
- histologically the appearance of the lung resembles a developing exocrine gland
- divisions of bronchi reach up to terminal bronchioles
- respiratory elements (respiratory bronchioles & alveoli) that are involved in respiration not formed
- fetus born during this period cannot survive

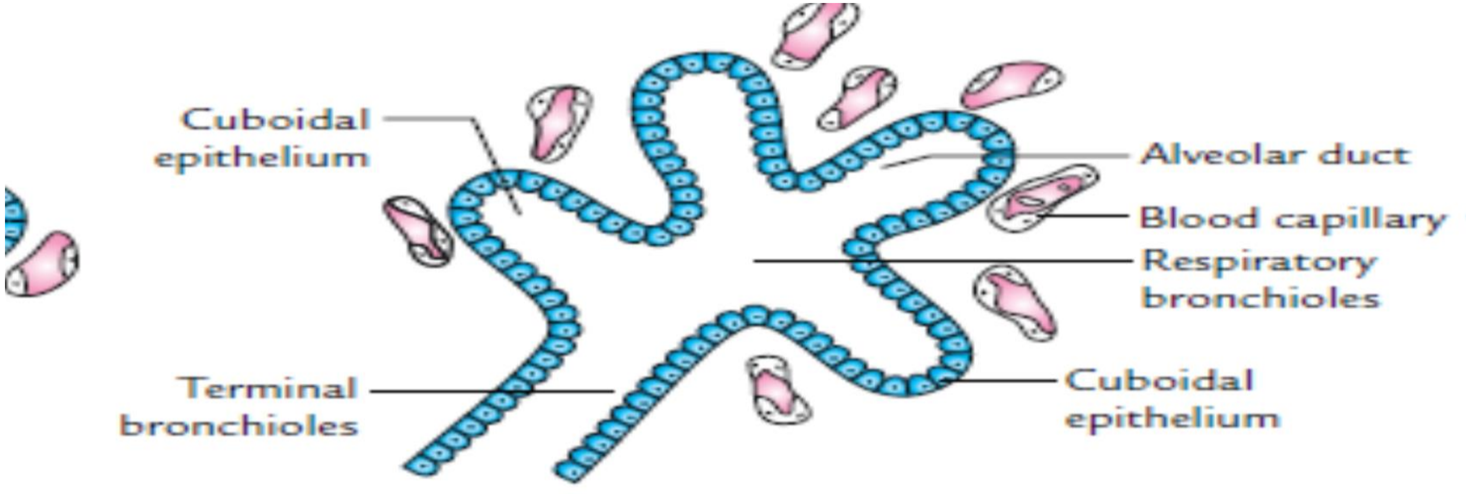


A Pseudoglandular stage (6–16 weeks)

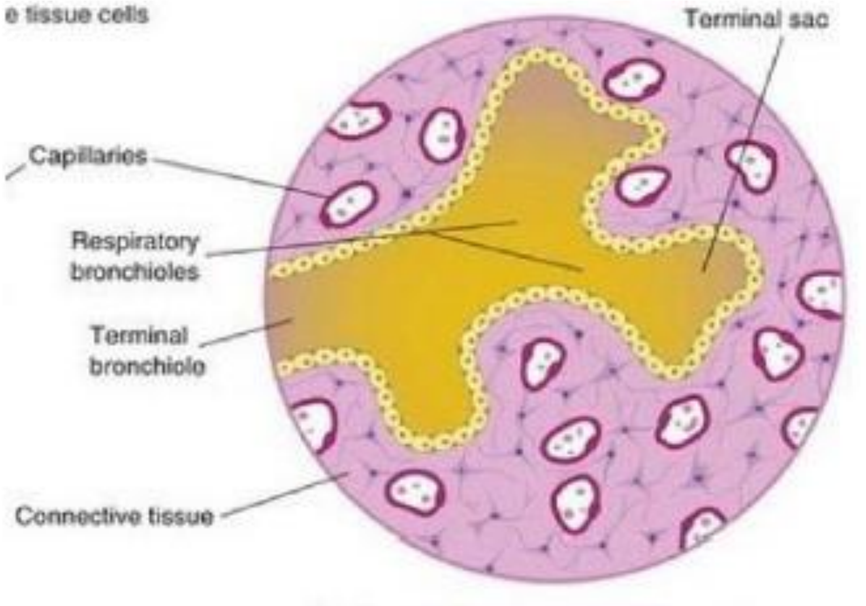


A Pseudoglandular stage

- **2. Canalicular stage (16–26 weeks):**
- lumens of terminal bronchioles dilate & subdivision of terminal bronchioles into respiratory bronchioles
- respiratory bronchioles divide into alveolar ducts
- Few terminal sacs (primitive alveoli) may also be formed at the ends of respiratory bronchioles
- fetus born towards the end of this period may survive if given intensive care
- lung tissue is well vascularized



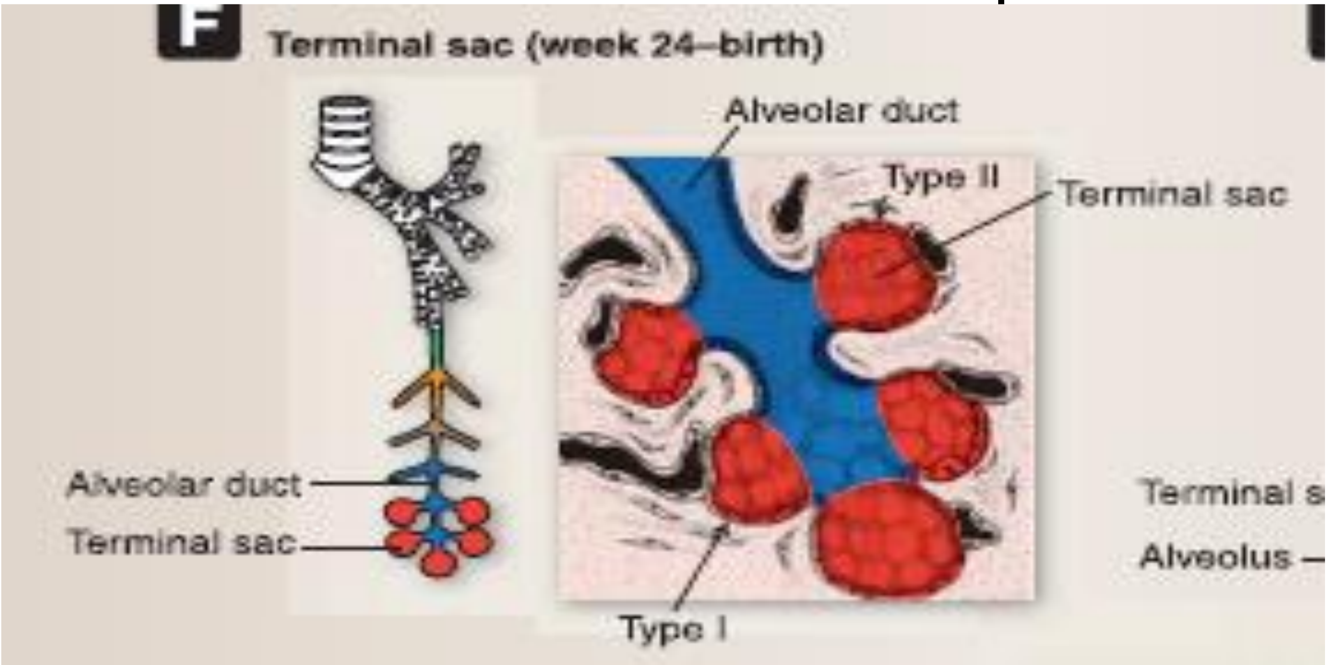
B Canalicular stage



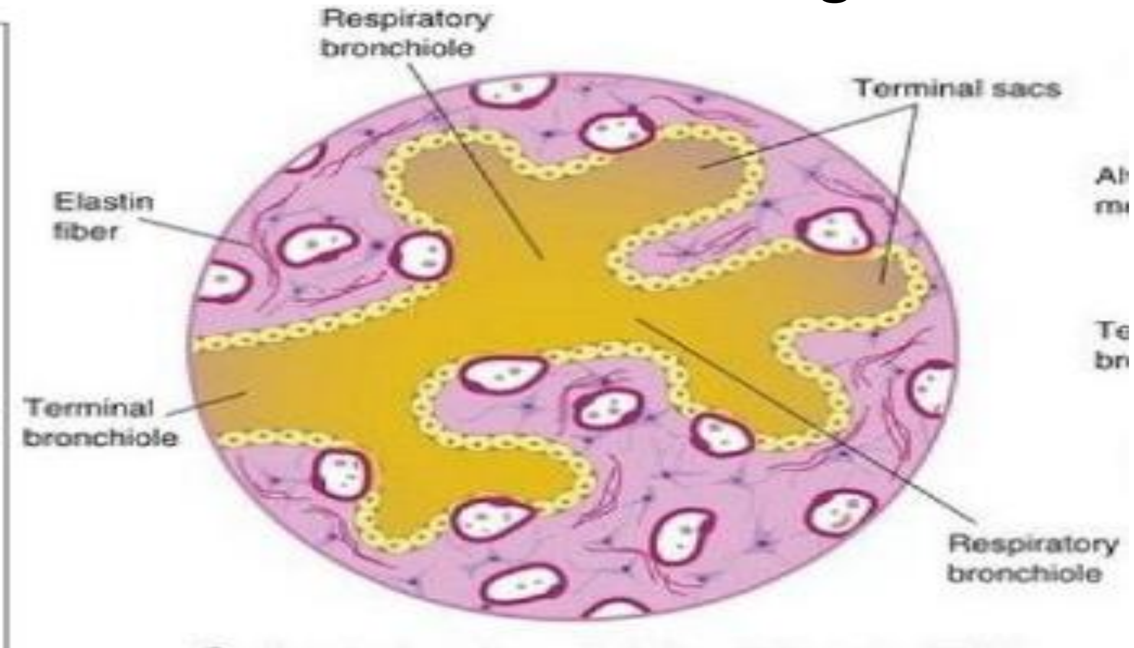
B Canalicular stage (16–26 weeks)

Canalicular period 16–26 wk

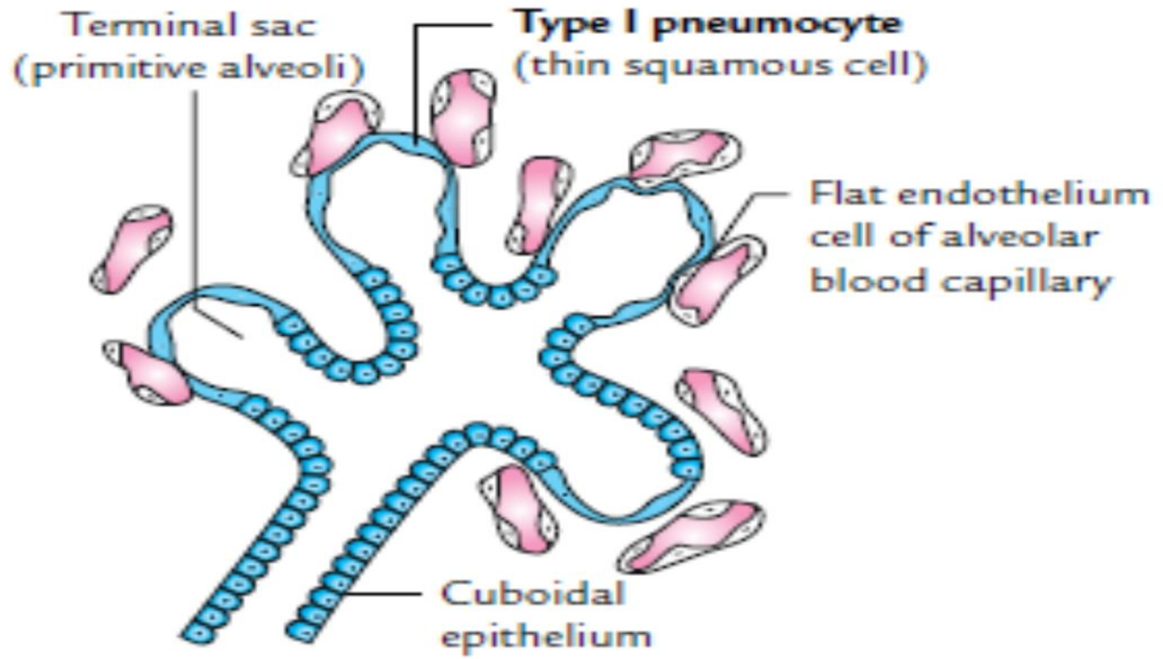
- **3. Terminal sac stage (26 weeks to birth):**
- large number of terminal sacs (primitive alveoli) develop
- Capillaries proliferate & form a plexus around the terminal sacs
- wall (epithelium) of terminal sacs becomes very thin & capillaries bulge into these sacs
- intimate contact between epithelial & endothelial cells establishes the blood–air barrier
- permits adequate exchange of gases for survival of the fetus if born prematurely



- Terminal sacs are lined by endodermal squamous cells - **type I alveolar epithelial cells** (type I pneumocytes) across which gaseous exchange takes place
- Scattered among type I pneumocytes, a few rounded type II alveolar epithelial cells (type II pneumocytes) are also present
- type II pneumocytes secrete surfactant & their number gradually increases towards the full-term
- lymphatic capillaries also increase in number
- The fetus born at this stage is viable

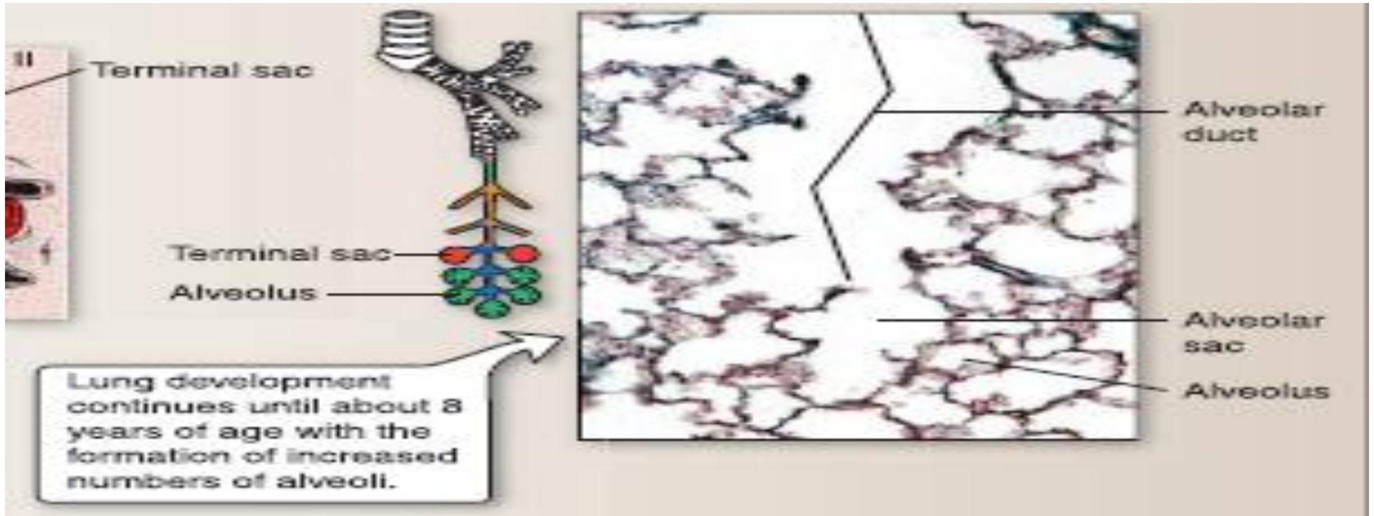


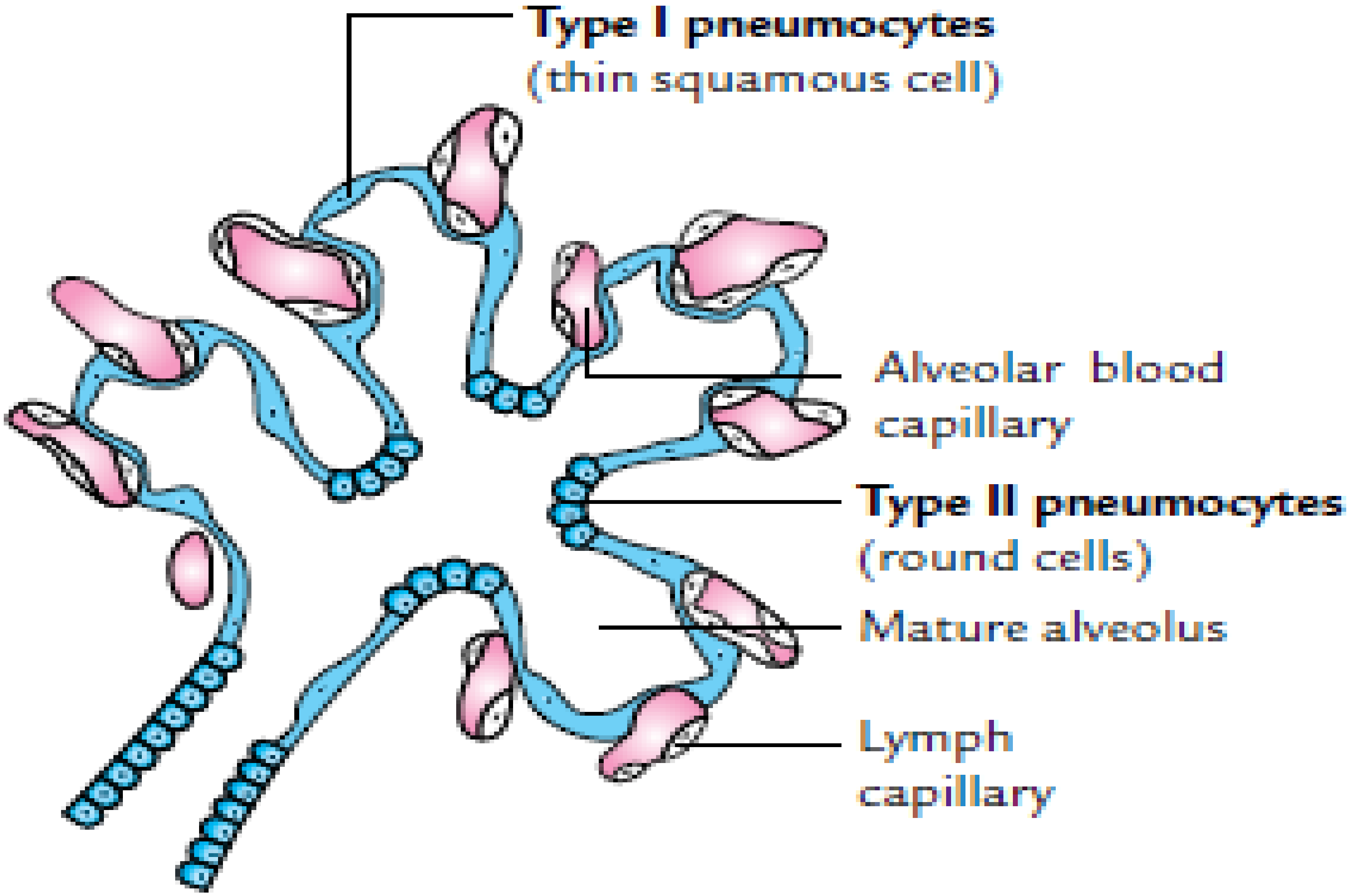
C Terminal sac (saccular) stage (26 weeks–birth)



C Saccular (terminal sac) stage

- **4. Alveolar stage (8 months to 8 years):**
- terminal sacs & respiratory bronchioles divide, form alveolar ducts
- at the end of alveolar ducts, definitive (true) alveoli are formed
- formation of mature (true) alveoli continues after birth till age of about 8 years
- true alveoli formed have extremely thin wall & are lined by type I & type II pneumocytes
- type II pneumocytes produce a sufficient amount of surfactant for survival
- capillaries also proliferate & vascularize the newly formed alveoli
- A number of alveoli reach the adult level by the 8th year





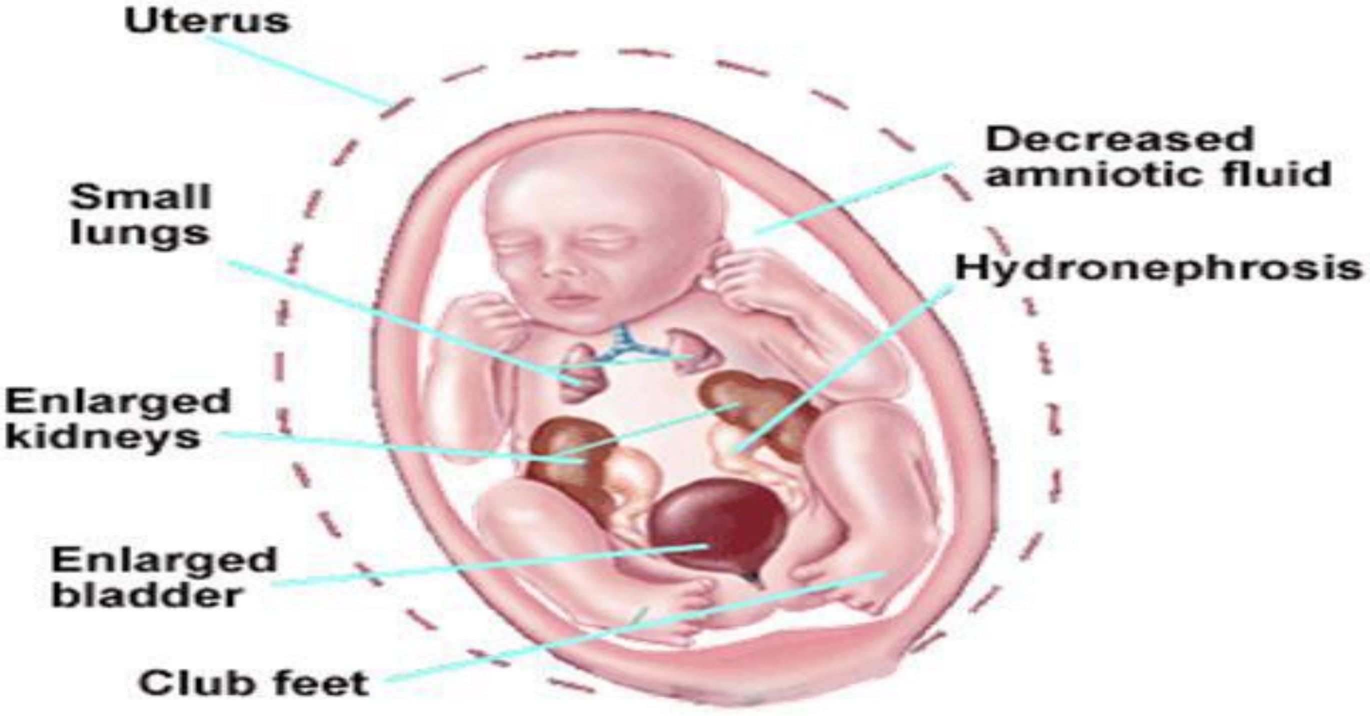
D

Alveolar stage

- **CLEARNCE OF FLUID IN THE LUNGS**
- At birth,
- lungs are approximately half-filled with fluid derived from amniotic cavity, lungs & tracheal glandular activity
- This fluid is cleared at birth by 3 routes*:*
- 1. Through mouth & nose by pressure on the fetal thorax during vaginal delivery
- 2. Into the pulmonary arteries, veins & capillaries
- 3. Into the lymphatics

- 3 factors are important for normal foetal lung development:
- 1. adequate thoracic space for lung growth
- 2. Foetal Breathing Movements
- 3. Adequate amniotic fluid volume
- Amniotic fluid volume could be diminished due to:
- a) Renal agenesis or insufficiency
- b) Placental insufficiency,
- c) amniotic fluid leakage, etc

- Oligohydramnios & Lung Hypoplasia
- lung development is retarded when oligohydramnios is chronic because of:-
- amniotic fluid leakage
- decreased production,
- Pulmonary hypoplasia results from restriction of foetal thorax
- This is the case in **Potter sequence**, etc



CLINICAL

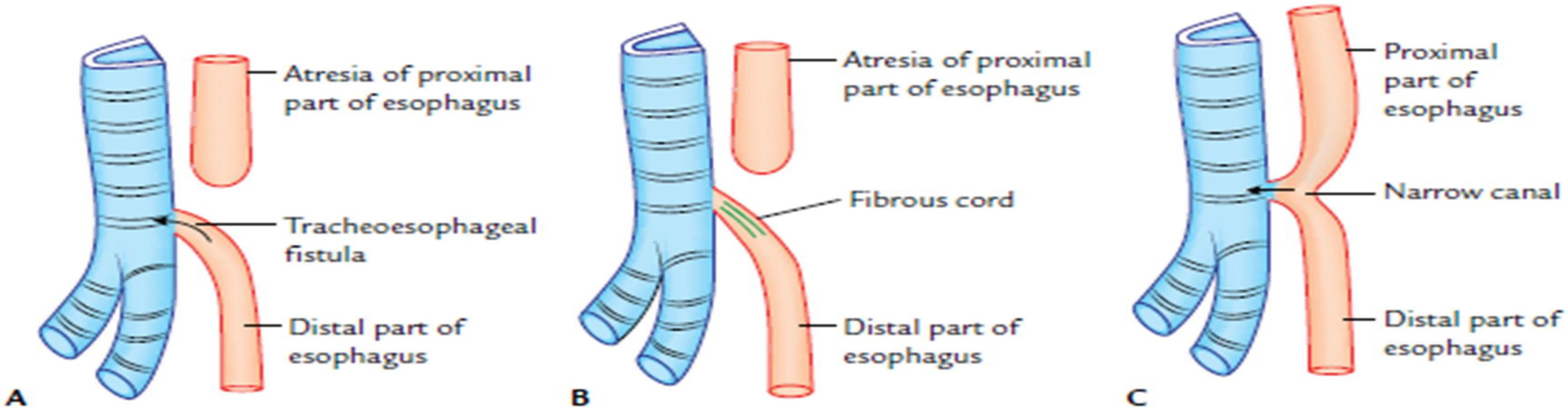
- **Remember Potter's syndrome**:
  - Renal agenesis
  - Oligohydramnios
  - Pulmonary hypoplasia
  - Deformed limbs
  - **Potter's facies**:
    - Flattened 'parrot-beaked' nose
    - Recessed chin and micrognathia
    - Prominent infraorbital folds
    - Low-set ears

- **Clinical Correlation**
- **Anomalies of larynx**
- 1. **Laryngeal atresia and stenosis:**
- rare anomaly results from failure of recanalization of the larynx that leads to obstruction of the upper respiratory tract (also called congenital high airway obstruction syndrome) due to narrowing of some sites
- Most commonly atresia (blockage) and stenosis (narrowing) is seen at the level of vocal folds

- 2. Laryngeal web:
- membranous, web-like tissue is present in the laryngeal lumen, usually at the level of vocal folds, which may partially obstruct the airway
- web-like tissue is derived from endodermal cells that fail to break out during recanalization of larynx

- **Anomalies of trachea**
- **1. Tracheoesophageal fistula (TEF):**
- an abnormal communication between the trachea & esophagus
- anomaly is often associated with esophageal atresia
- occurs in 1/3000–4500 births
- TEF occurs due to defective development of tracheoesophageal septum
- tracheoesophageal folds fail to fuse with each other in a small segment, producing an abnormal communication between the trachea & esophagus—the TEF

- The various types of TEF are:
- (a) Upper part of the esophagus ends in a blind pouch & lower part communicates with the trachea (85–90%)
- (b) As type (a) but the tracheoesophageal communication/canal is replaced by a fibrous cord
- (c) Both upper & lower parts of the esophagus communicate with the trachea by a common narrow canal
- It is called H-shaped TEF



- (d) Upper part of the esophagus communicates with the trachea & lower end forms a blind pouch
- (e) Both upper & lower parts of the esophagus communicate with the trachea separately
- When milk or fluid is given to newborn infants with TEF, there occurs coughing & choking because milk or fluid enters into the respiratory tract
- may also lead to lung infection (pneumonia)

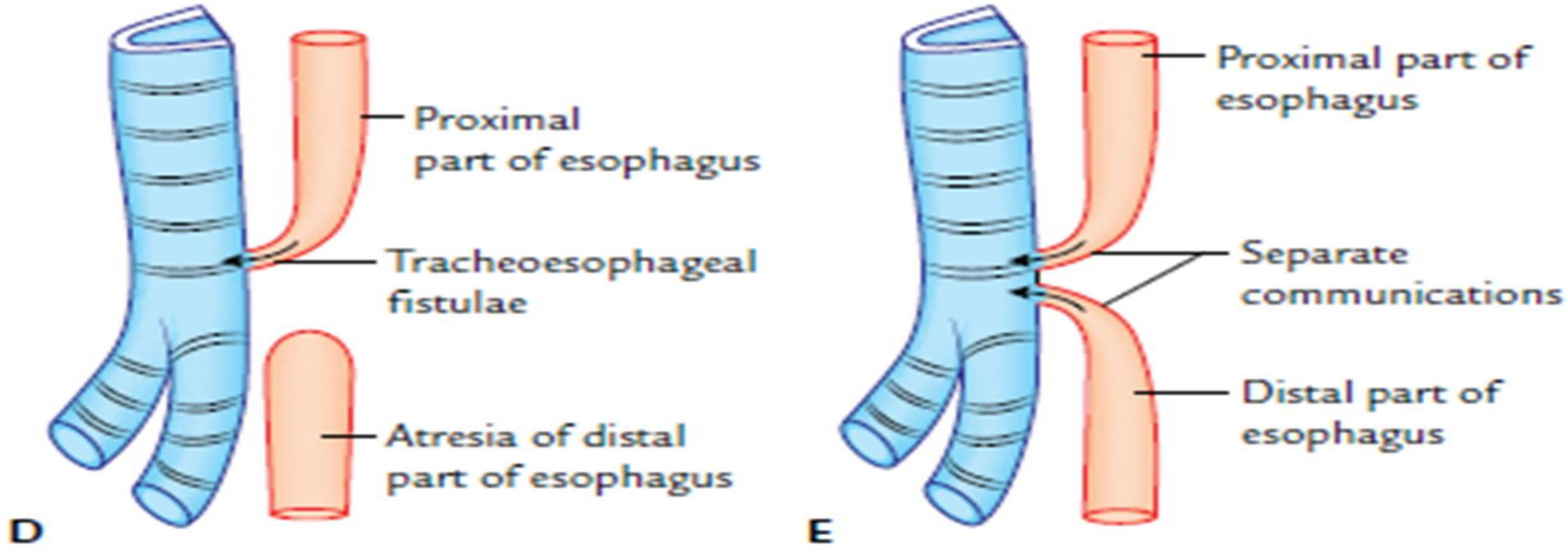


Fig. 16.5 Types of tracheoesophageal fistulae. A. Atresia of the esophagus and tracheoesophageal fistula. B. Atresia of the esophagus and connection between the distal part of esophagus and trachea by a fibrous cord. C. Both proximal and distal parts of esophagus connected to the trachea by a narrow canal. D. Atresia of distal part of esophagus and connection between the proximal part of esophagus and trachea. E. Separate communications of proximal and distal parts of esophagus to the trachea.

- **2. Tracheal stenosis (narrowing of trachea):**
- rare and occurs due to anterior deviation of the tracheoesophageal septum
- **3. Tracheal atresia (tracheal obstruction):**
- occurs due to the presence of a web of tissue within tracheal lumen
- tissue is derived from proliferation of endodermal cells
- **4. Tracheal bronchus and tracheal lobe:**
- Sometimes the trachea presents a diverticulum that may either end blindly
- (blind bronchus) or supply a lobe of lung called tracheal lobe, which is not a normal part of the lung

- Sometimes it may replace a normal bronchus, viz., apical bronchus of upper lobe of lung

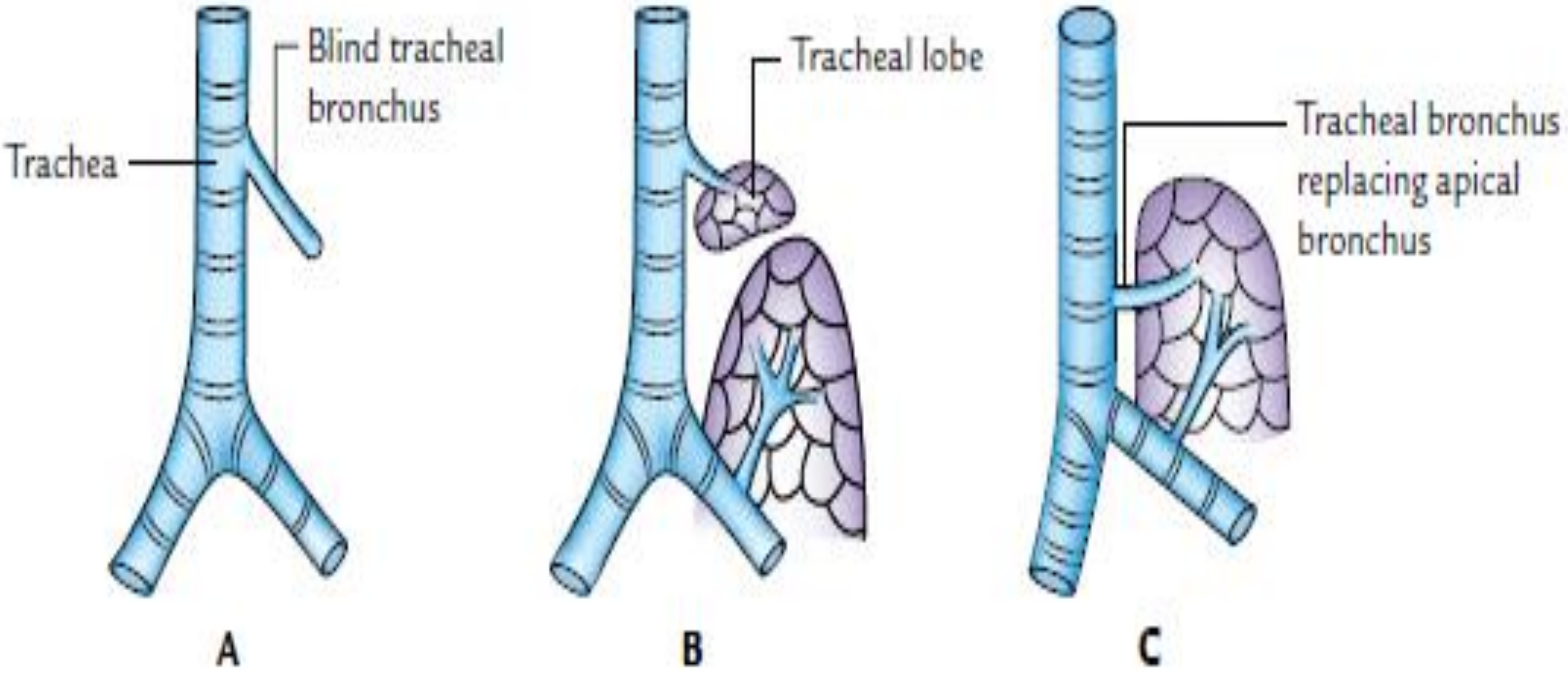
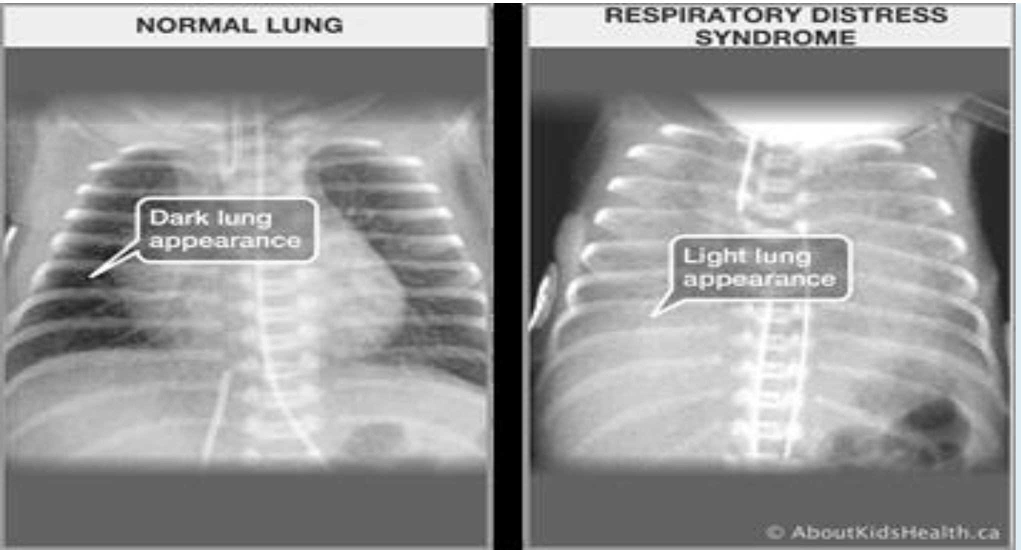


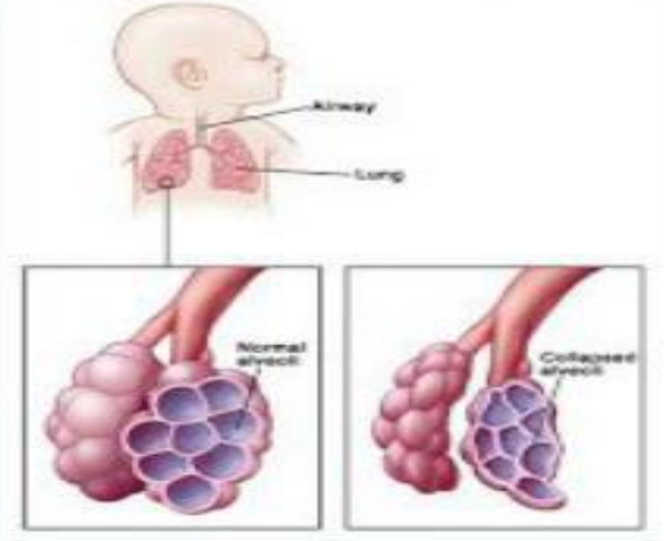
Fig. 16.6 Accessory bronchi arising from trachea. A. Blind tracheal bronchus. B. Tracheal bronchus supplying an accessory mass of lung tissue. C. Tracheal bronchus replacing apical bronchus.

- **1. Medicolegal importance of the newborn lung:**
- lungs of a newborn infant born alive always contain some air within it hence they float in water & crepitate on pressure
- lungs of a newborn infant born dead (stillborn) or diseased ones are firm & sink when placed in water because they contain fluid, not air
- it is firm and does not crepitate on pressure
- This fact is of medicolegal significance, it tells whether the newborn infant was born alive or born dead (stillborn)

- **2. Respiratory distress syndrome (RDS)/** hyaline membrane disease **:**
- affects about 2% of live newborn infants
- infants with this condition develop rapid, labored breathing shortly after birth
- common in premature infants
- common cause is deficiency of surfactant
- alveoli of lungs are often filled with fluid having high protein content, which resembles glassy hyaline membrane
- corticosteroids (glucocorticoids) & thyroxine, which are involved in lung maturation & production of surfactant, may be used therapeutically



- Also called hyaline membrane disease.
- Most common cause of respiratory distress in preterm infants.
- Due to structural and functional immaturity of lungs.
  - Underdeveloped parenchyma
  - Surfactant deficiency
    - Type II pneumatocytes
- Results in decreased lung compliance, unstable alveoli



- **3. Congenital anomalies of the lung**
- **(a) Agenesis (nondevelopment) of the lung:**
- rare & occurs if one of the bronchial buds fails to develop
- Agenesis of both the lungs is still rare
- unilateral agenesis of the lung is compatible with life
- lung hypoplasia is characterized by a markedly reduced lung volume and hypertrophy of smooth muscle of pulmonary arteries

- **(c) Abnormal lobes of lungs:**
- right lung may consist of 2 lobes instead of 3 or left lung may consist of three lobes instead of 2
- anomalies occur due to abnormal division of principal bronchi into lobar bronchi, and are associated with anomalies of lung fissures
- anomalies are not clinically significant

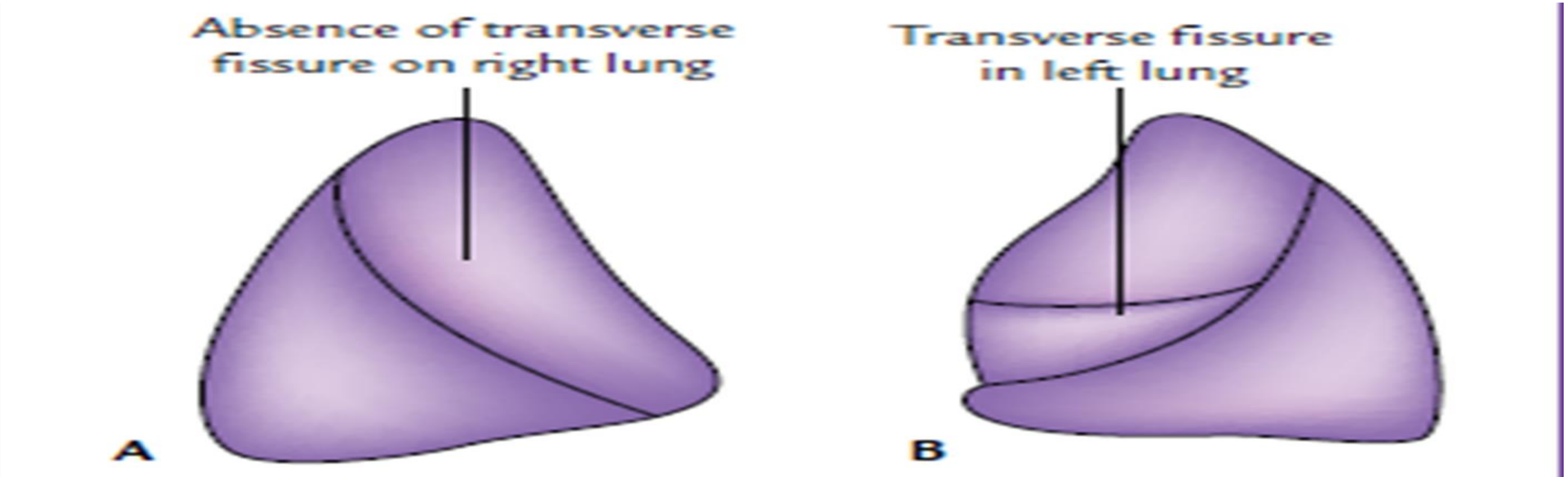


Fig. 16.9 Abnormal lobes of the lungs. A. Right lung with two lobes due to the absence of transverse fissure. B. Left lung with three lobes due to the presence of the transverse fissure.

- **(d) Azygos lobe of the lung (Lobe of Wrisberg):**
- a portion of upper lobe of right lung that lies medial to arch of azygos vein, commonest accessory lobe of the lung
- azygos vein lies in floor of vertical fissure , which produces a linear marking on a radiograph of the lungs
- azygos lobe is found in the right lung because azygos vein itself is on the right side

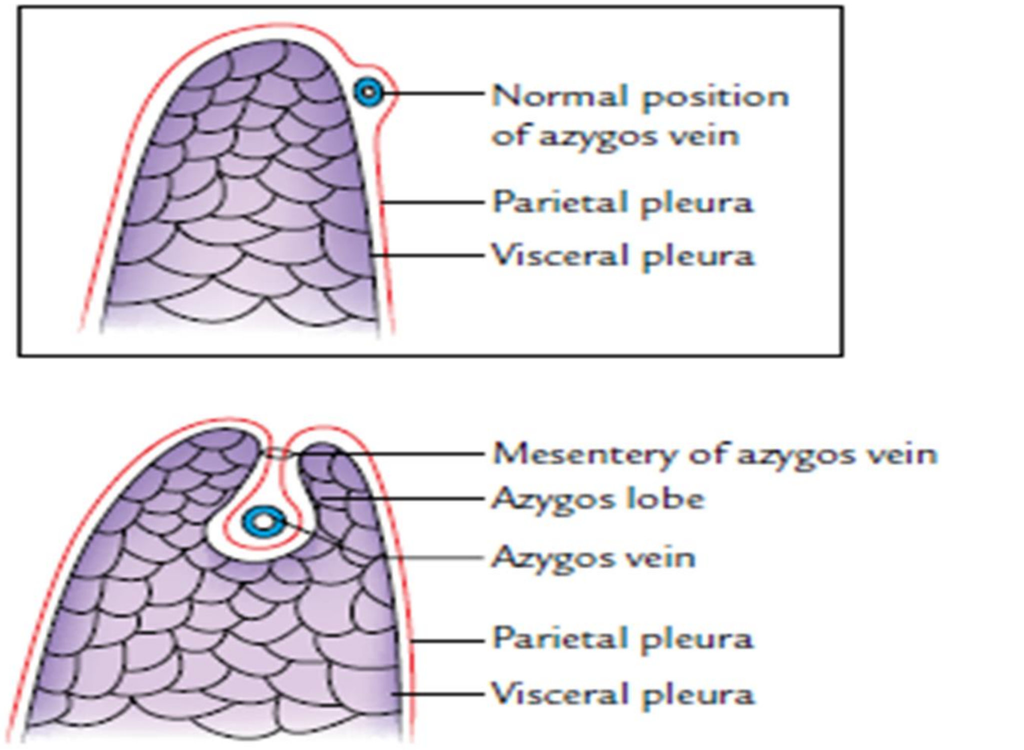
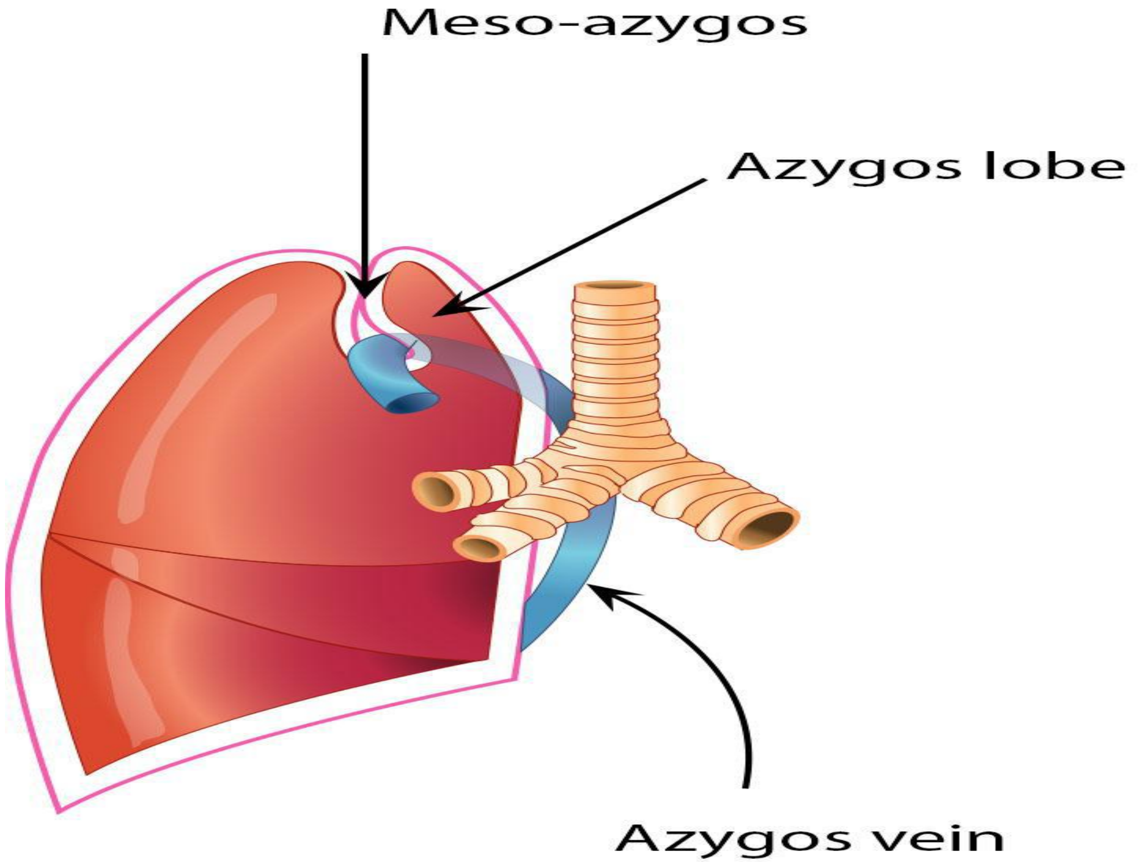


Fig. 16.10 Azygos lobe of the lung. Figure in the inset shows normal relationship of azygos vein with the lung.



- **(e) Ectopic lung lobes:**
- arise from trachea or esophagus
- These lobes probably form additional respiratory buds of the trachea and foregut that develop independently of main respiratory system
- Almost always located at the base of the left lung and Is nonfunctional

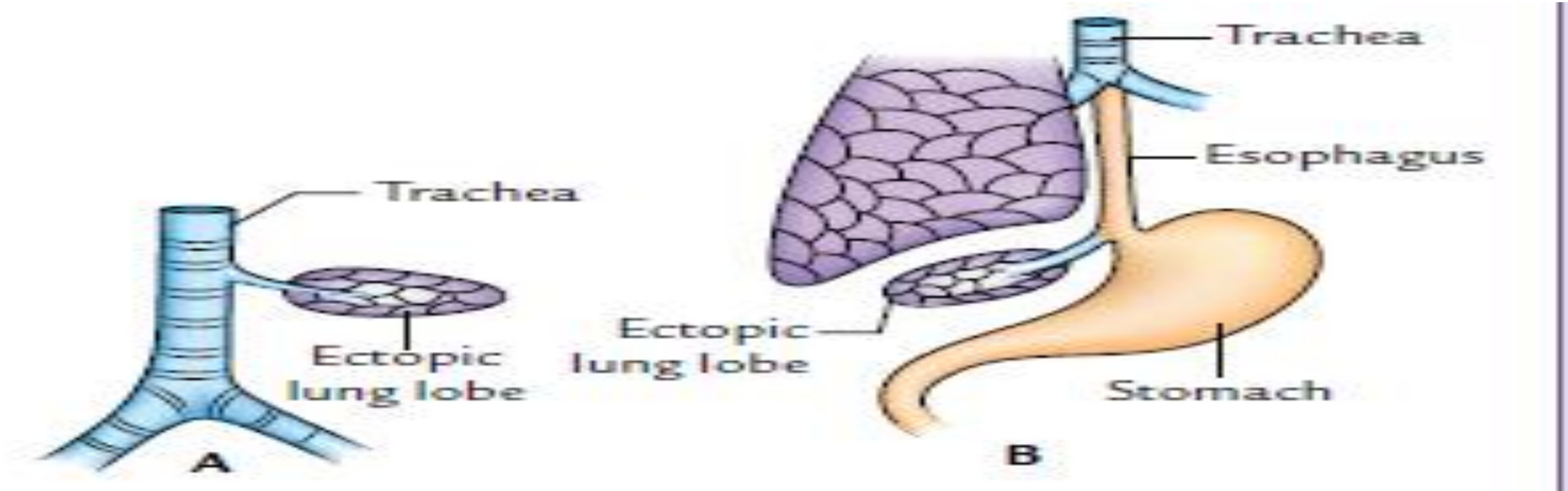


Fig. 16.11 Ectopic lung lobes. A. Arising from trachea. B. Arising from esophagus.

- **(f) Congenital polycystic lung:**
- Numerous cysts (air-filled or fluid-filled)are thought to be formed in the lung due to abnormal dilatation of terminal bronchioles
- give the lung a honeycomb appearance, which can be seen in the radiograph
- cysts drain poorly, frequently cause chronic infections
- Congenital lung cysts are found in the periphery of the lungs
- For this reason, congenital polycystic lung is clinically most important